

수위 예측 AI 성능 개선 결과 1

이전 학습 결과

학습 데이터

learning_rate = 0.0005

강우 이벤트

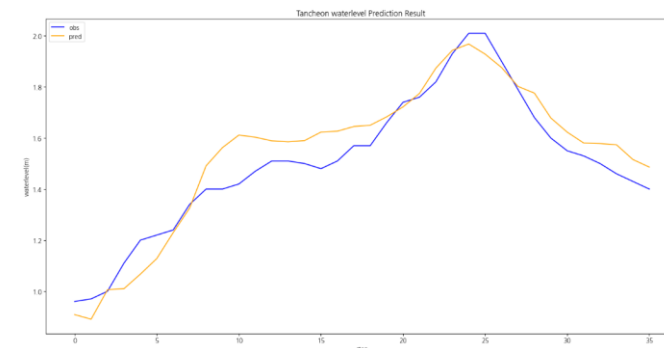
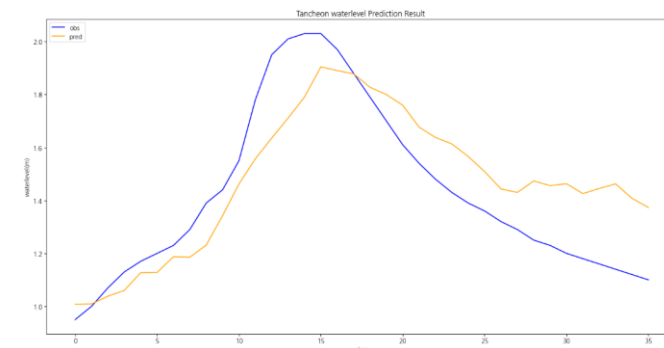
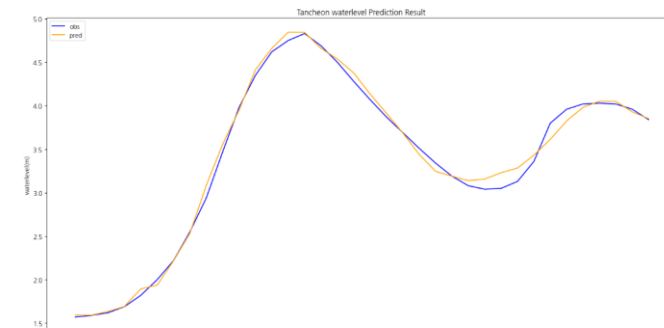
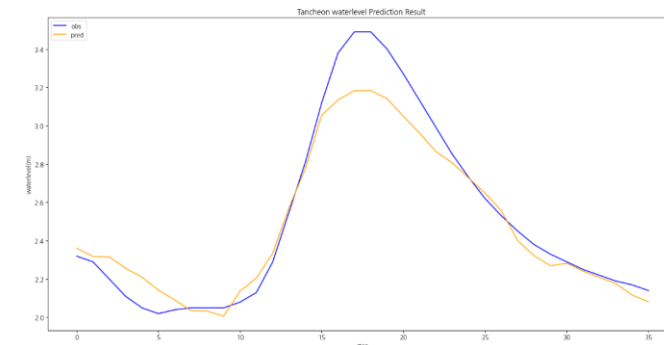
time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 6개 (140820, 170715, 200805, 220629, 220807, 230713) 사용

검증 : 학습 데이터에서 사용한 2개 데이터 (220629, 220807) + 학습에 사용되지 않은(250716, 250814) 사용

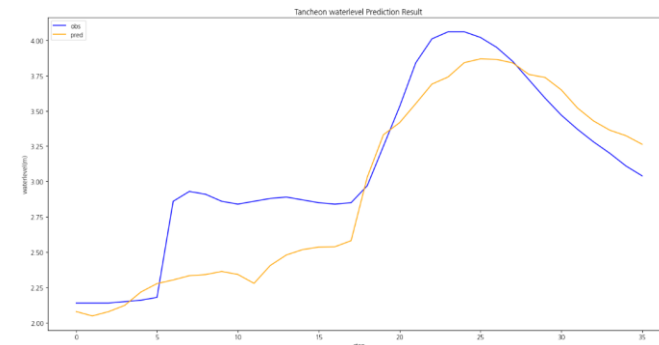
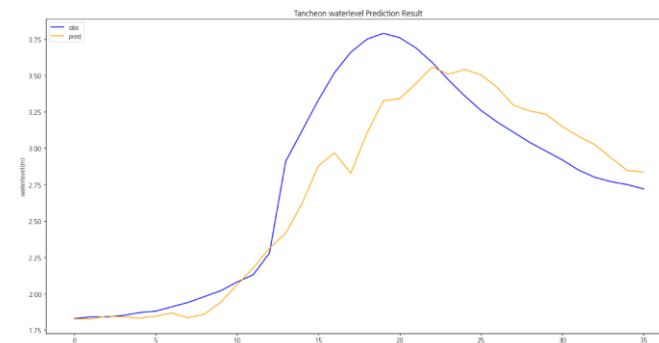
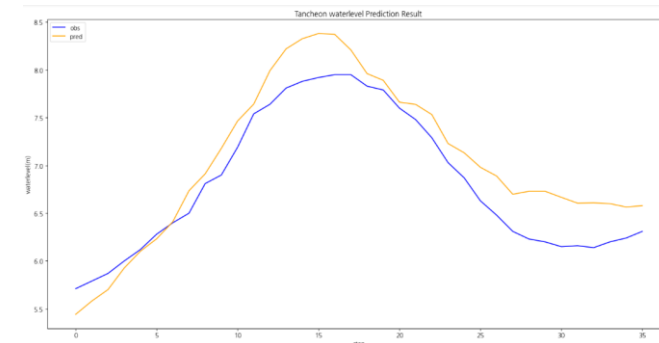
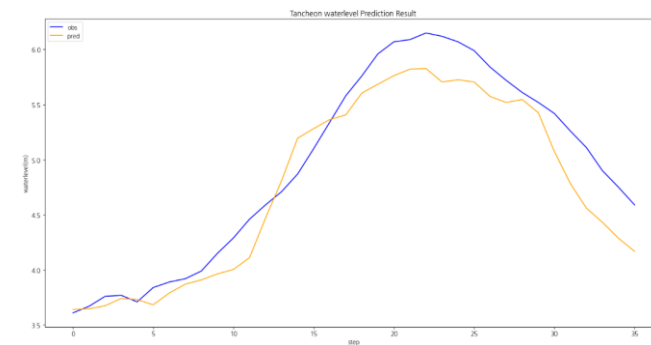
수위예측 smoothing = 1

	궁내교	대곡교
testdata_220630	r2 : 0.9334831558683871 mse : 0.014662755503537002 mae : 0.08591636021931968 rmse : 0.12108986540390984	r2 : 0.8461369366287653 mse : 0.11212509555879795 mae : 0.28038430240419177 rmse : 0.33485085569369233
testdata_220808	r2 : 0.9933156779676867 mse : 0.0061528580861704455 mae : 0.05965378244717913 rmse : 0.07844015608201227	r2 : 0.7864821009981524 mse : 0.11192937768607407 mae : 0.2791808260811699 rmse : 0.33455848171294966
testdata_250716	r2 : 0.6786097511859852 mse : 0.03224258396157489 mae : 0.15377818557951184 rmse : 0.17956220081513505	r2 : 0.7787861120789699 mse : 0.10215350102682166 mae : 0.23608912044101293 rmse : 0.319614613287349
testdata_250814	r2 : 0.8931957393452244 mse : 0.007529922885038036 mae : 0.07411362230777739 rmse : 0.08677512826287286	r2 : 0.6946340343597268 mse : 0.10599222036528559 mae : 0.2742249170939128 rmse : 0.32556446422373186



수위예측 smoothing = 2

궁내교		대곡교	
testdata_220630	r2 : 0.7088730596475936 mse : 0.06417507027894874 mae : 0.19718893157111278 rmse : 0.2533279895292834	r2 : 0.9004934548421246 mse : 0.07251370562948573 mae : 0.22338981760872734 rmse : 0.2692836898690408	
testdata_220808	r2 : 0.9773663276419933 mse : 0.020834091073780862 mae : 0.10682662063174778 rmse : 0.1443401921634472	r2 : 0.8118272173611429 mse : 0.09864307656025555 mae : 0.2750794050428602 rmse : 0.3140749537296082	
testdata_250716	r2 : 0.5100765623979442 mse : 0.049150207978988454 mae : 0.16626809497674305 rmse : 0.22169846183270747	r2 : 0.8130232584048014 mse : 0.08634326236947386 mae : 0.21145669579505919 rmse : 0.29384224061471126	
testdata_250814	r2 : 0.8737931172772833 mse : 0.008897848162957201 mae : 0.06456567585468291 rmse : 0.09432840591760894	r2 : 0.7407194444433576 mse : 0.08999602075290633 mae : 0.2423573001225789 rmse : 0.29999336784820146	



학습 데이터 증강

MIET

MIET 학습 데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 강우 포함 파일 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+ 기존 학습 데이터 통합 후 중복 이벤트 제거

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	서울시 (대곡교)_WL	대곡교_Ti	궁내교_Ti
2014-08-20 10:50	0.0	0.0	0.0	0.0	0.0	0.0	0.96	1.84	0.0	0.0

학습 : 총 67 개 데이터 사용

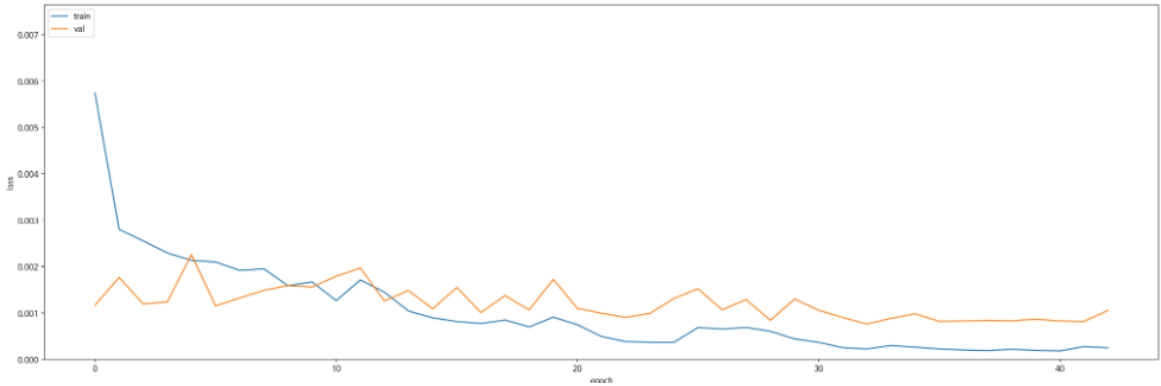
검증 : 학습 데이터에 포함된 4개 데이터 (220630, 220808, 250716, 250814) 사용

loss_function : mse

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5330 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0479	0.0308	0.0737	0.983723	0.984563	3.499491

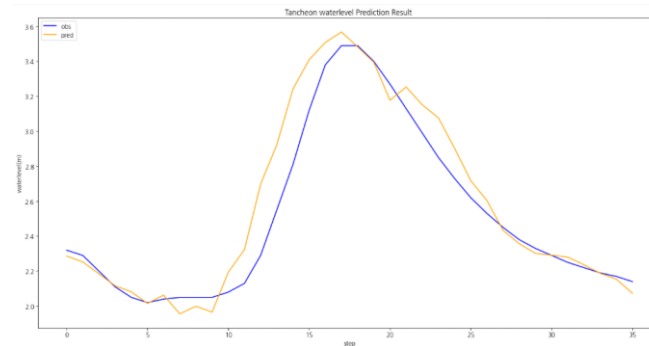
Valid (1219 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0790	0.0562	0.1265	0.900232	0.903904	21.429995

Test (594 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0583	0.0435	0.0896	0.931545	0.934990	3.399140

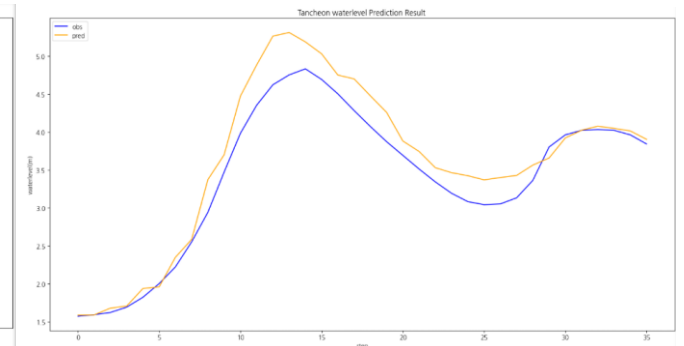
loss_function : mse

예측코드 - 궁내교

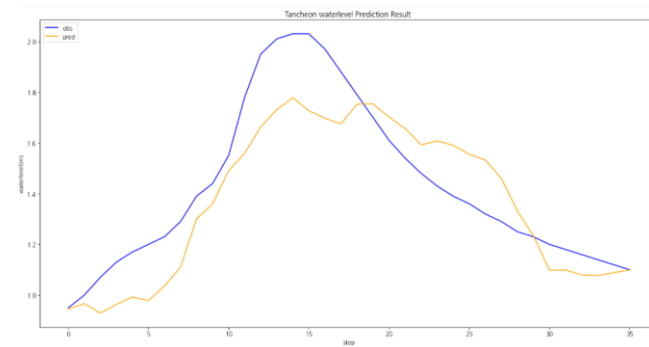
	궁내교
testdata_220630	r2 : 0.8961080783037266 mse : 0.022901595325402445 mae : 0.09876218146748011 rmse : 0.15133273051591464
testdata_220808	r2 : 0.9092116546241392 mse : 0.08356985230147296 mae : 0.2276965120103624 rmse : 0.2890845071972432
testdata_250716	r2 : 0.7353645568784295 mse : 0.02654881573271844 mae : 0.1382774554358588 rmse : 0.16293807330614427
testdata_250814	r2 : 0.6692581867518503 mse : 0.023317986879438816 mae : 0.11295003725422752 rmse : 0.1527022818409693



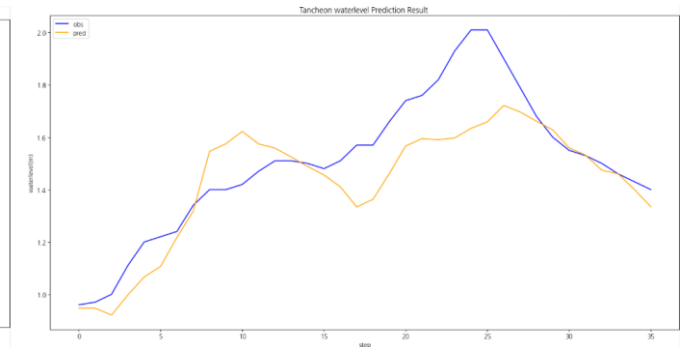
testdata_220630



testdata_220808



testdata_250716



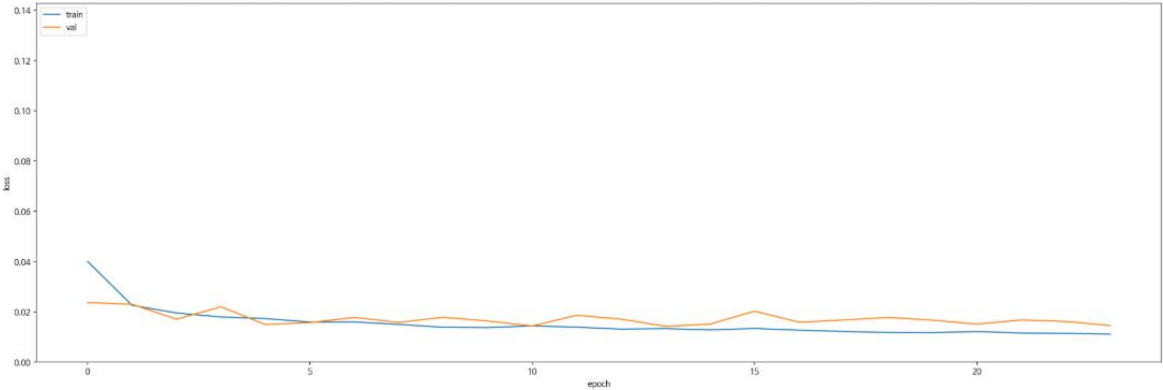
testdata_250814

loss_function : mae

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5401건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0617	0.0340	0.1814	0.901763	0.911266	0.440397

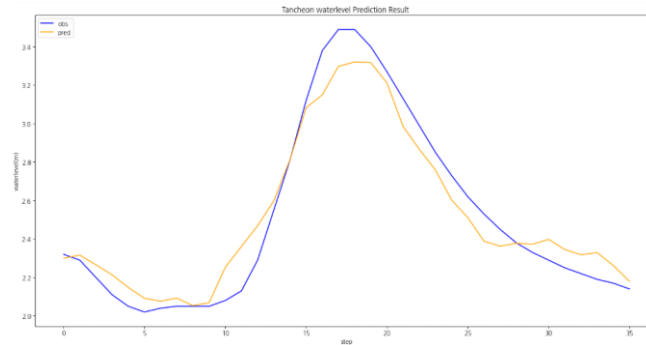
Valid (1318건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0653	0.0469	0.0992	0.934844	0.938947	3.080681

Test (693건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0513	0.0366	0.0875	0.924847	0.940583	2.353045

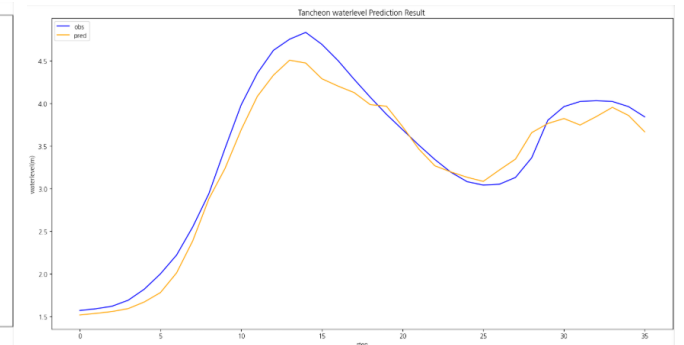
loss_function : mae

예측코드 - 궁내교

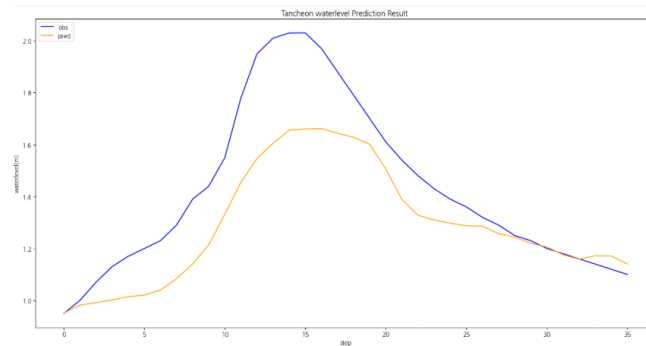
	궁내교
testdata_220630	r2 : 0.9440788069853889 mse : 0.012327084836089656 mae : 0.0924820923805237 rmse : 0.11102740578834425
testdata_220808	r2 : 0.9608196777555482 mse : 0.03606513291477533 mae : 0.15904683166080053 rmse : 0.18990822234641483
testdata_250716	r2 : 0.6380790141928143 mse : 0.03630871756503421 mae : 0.1456537667910258 rmse : 0.19054846513429124
testdata_250814	r2 : 0.5275173512163556 mse : 0.03331101107809856 mae : 0.13911650240421297 rmse : 0.1825130435834616



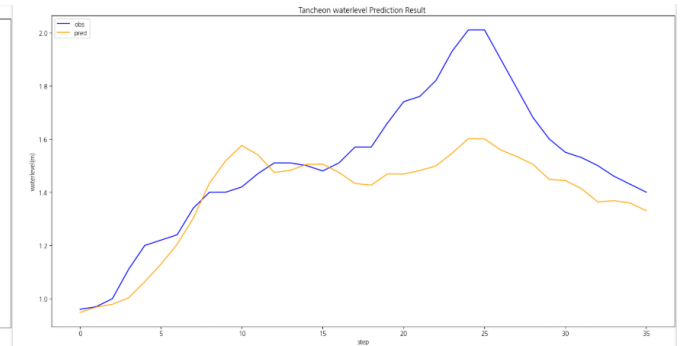
testdata_220630



testdata_220808



testdata_250716



testdata_250814

IETD

IETD 학습 데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005

홍수통제소 API 호출 → IETD 활용 강우 이벤트 추출 → 관심 강우 포함 파일 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+ 기존 학습 데이터 통합 후 중복 이벤트 제거

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	서울시 (대곡교)_WL	대곡교_Ti	궁내교_Ti
2014-08-20 10:50	0.0	0.0	0.0	0.0	0.0	0.0	0.96	1.84	0.0	0.0

학습 : 총 41 개 데이터 사용

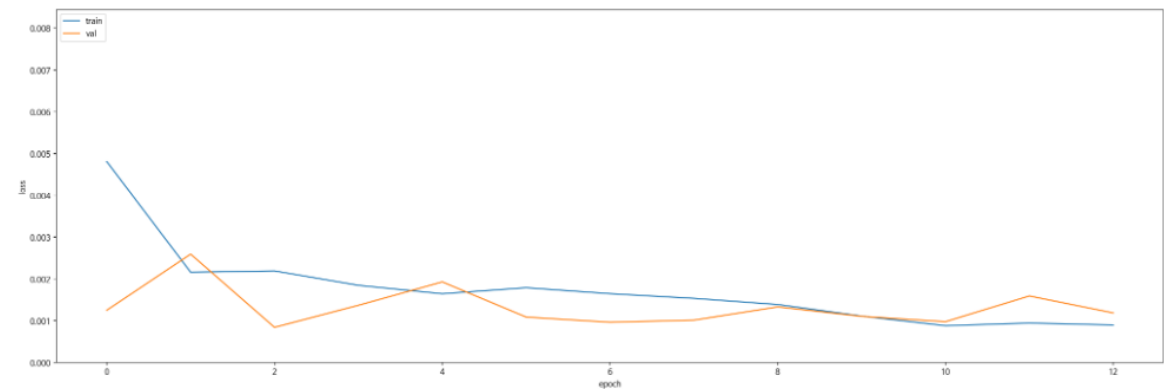
검증 : 학습 데이터에 포함된 4개 데이터 (220630, 220808, 250716, 250814) 사용

loss_function : mse

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (6568 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0963	0.0585	0.1893	0.877040	0.886103	1.200072

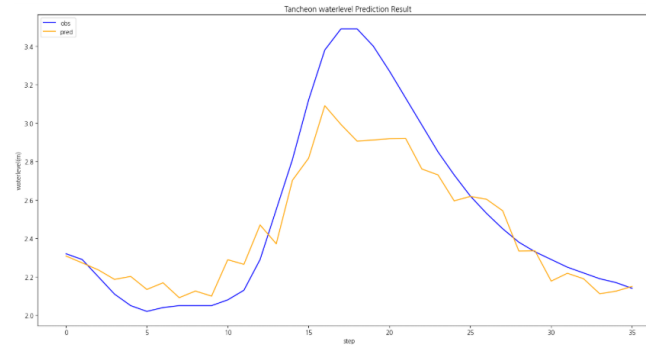
Valid (1546 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0915	0.0706	0.1316	0.883410	0.885025	6.182686

Test (591 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0936	0.0751	0.1309	0.828633	0.848889	1.180762

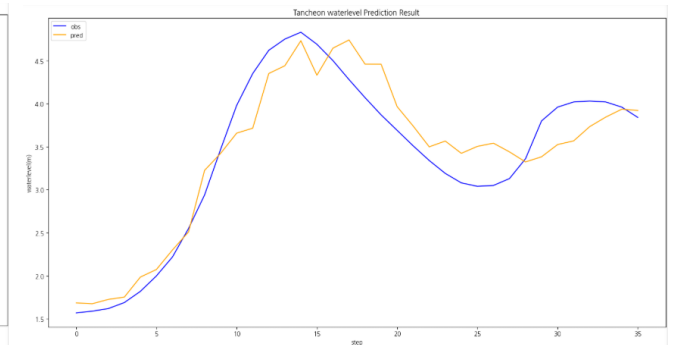
loss_function : mse

예측코드 - 궁내교

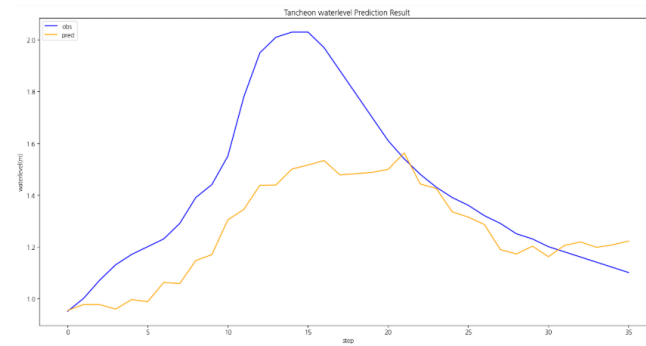
	궁내교
testdata_220630	r2 : 0.8107090885336524 mse : 0.04172666923856095 mae : 0.14577401796976724 rmse : 0.20427106804087786
testdata_220808	r2 : 0.8983921108487866 mse : 0.09352914467024825 mae : 0.2558622750971053 rmse : 0.3058253499470707
testdata_250716	r2 : 0.3699167390762792 mse : 0.06321135292089193 mae : 0.18525710814528992 rmse : 0.25141868053287514
testdata_250814	r2 : 0.38497702240945686 mse : 0.043360401218003254 mae : 0.155895366470019 rmse : 0.20823160475298474



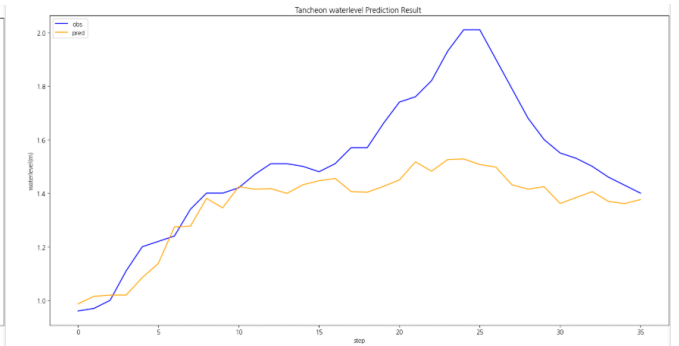
testdata_220630



testdata_220808



testdata_250716



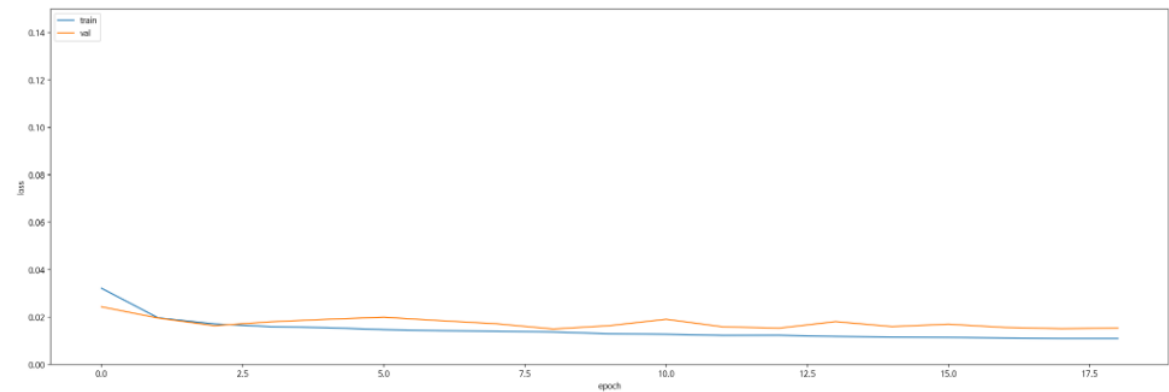
testdata_250814

loss_function : mae

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (6568 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0581	0.0333	0.1691	0.901961	0.903945	0.810389

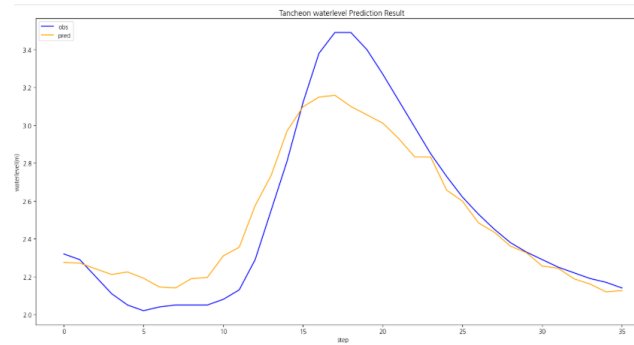
Valid (1546 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0678	0.0541	0.0964	0.937339	0.942005	3.439158

Test (591 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0691	0.0559	0.0984	0.903217	0.921347	4.550139

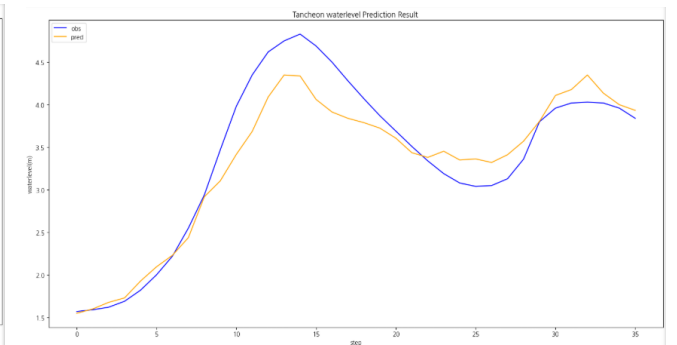
loss_function : mae

예측코드 - 궁내교

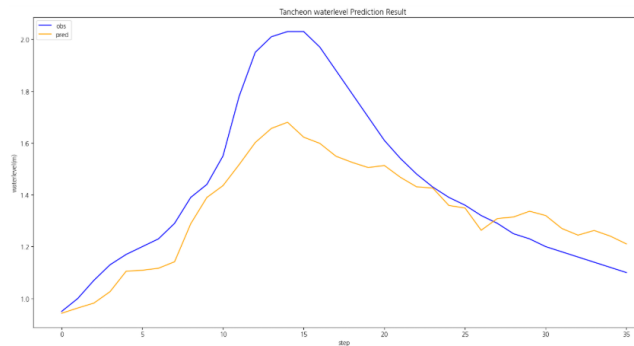
	궁내교
testdata_220630	r2 : 0.8788828603652485 mse : 0.026698666013652478 mae : 0.12266025543212894 rmse : 0.16339726440076185
testdata_220808	r2 : 0.9010840174237275 mse : 0.09105126896994951 mae : 0.22974014374944898 rmse : 0.3017470281045855
testdata_250716	r2 : 0.6783058984645005 mse : 0.0322730671418225 mae : 0.13786608695983885 rmse : 0.1796470627141521
testdata_250814	r2 : 0.5365595534008036 mse : 0.032673516986173747 mae : 0.12506023744742073 rmse : 0.1807581726677213



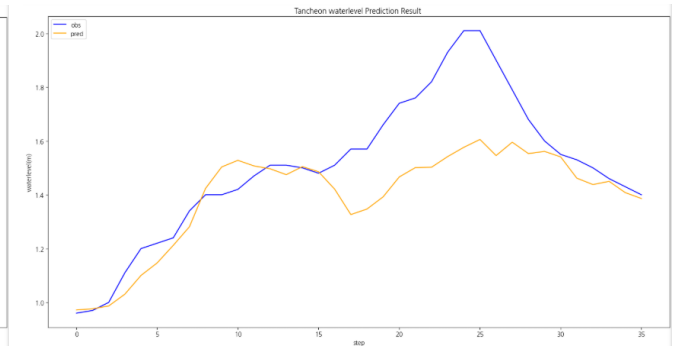
testdata_220630



testdata_220808



testdata_250716



testdata_250814

예측에 사용할 데이터
학습에서 제외

MIET

MIET 학습 데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 강우 포함 파일 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+ 기존 학습 데이터 통합 후 중복 이벤트 제거

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	서울시 (대곡교)_WL	대곡교_Ti	궁내교_Ti
2014-08-20 10:50	0.0	0.0	0.0	0.0	0.0	0.0	0.96	1.84	0.0	0.0

학습 : 총 67개 中 63개 데이터 사용

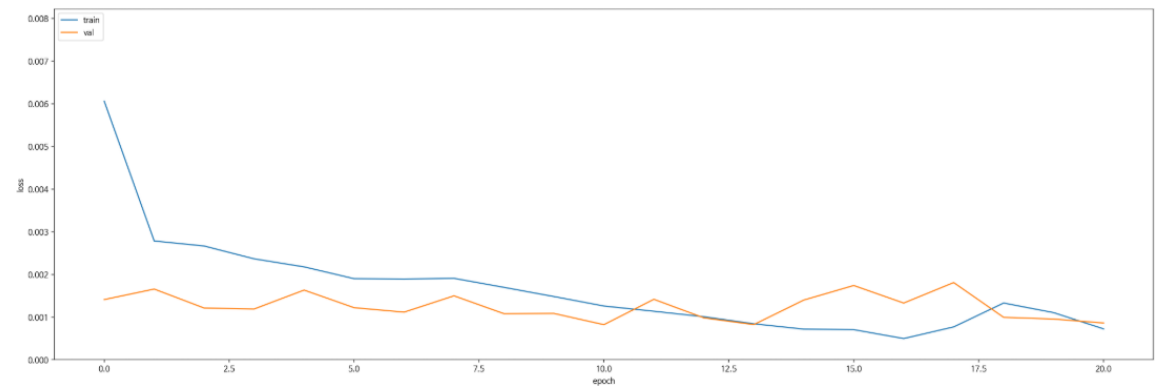
검증 : 학습 데이터에서 제외한 3개 데이터 (170723, 220713, 230711) 사용

loss_function : mse

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5101 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0842	0.0509	0.1492	0.935564	0.939824	21.222795

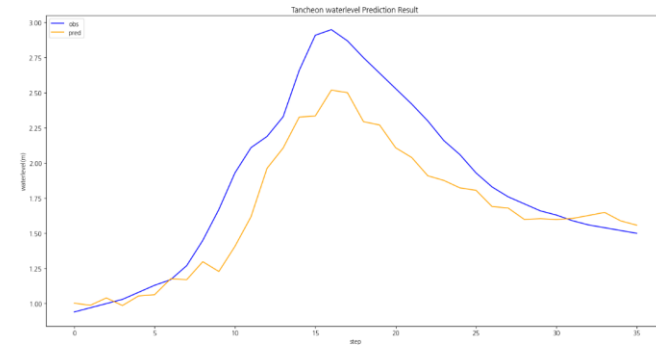
Valid (1247 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0929	0.0708	0.1313	0.889257	0.891299	27.523746

Test (693 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0711	0.0560	0.1020	0.897936	0.913832	7.785929

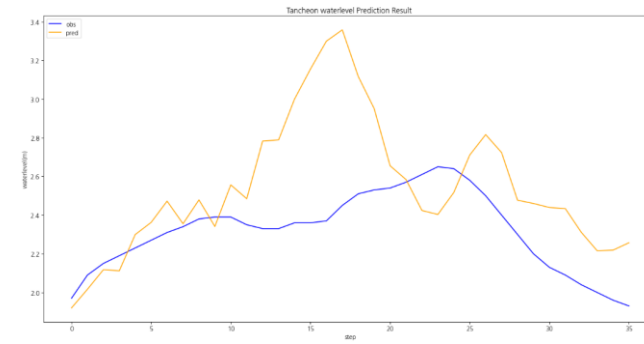
loss_function : mse

예측코드 - 궁내교

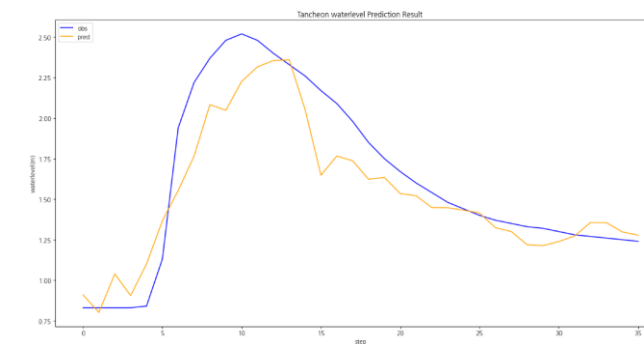
	궁내교
testdata_170723	r2 : 0.7911127456690144 mse : 0.07375852468605092 mae : 0.20897888594203526 rmse : 0.2715852070456911
testdata_220712	r2 : -2.3285858847672483 mse : 0.13207294573268763 mae : 0.27360212643941245 rmse : 0.3634184168870472
testdata_230711	r2 : 0.8460995310832496 mse : 0.043067656847119315 mae : 0.15631436467170717 rmse : 0.2075274845583575



testdata_170723



testdata_220712



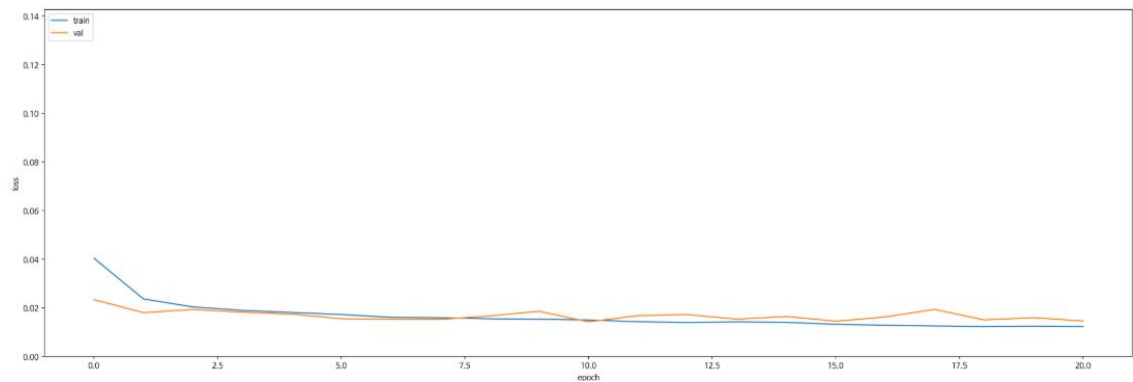
testdata_230711

loss_function : mae

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5101 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0654	0.0354	0.1901	0.895468	0.895784	3.965477

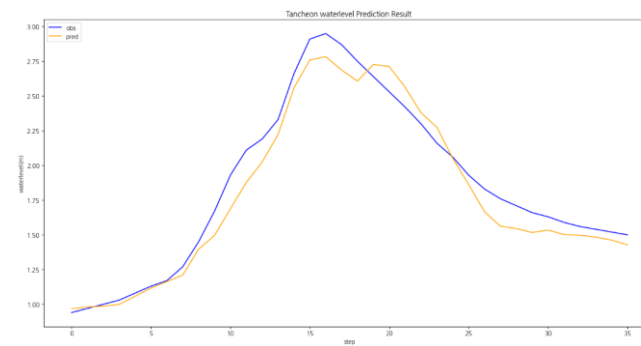
Valid (1247 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0653	0.0502	0.0903	0.947582	0.950634	12.819827

Test (693 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0531	0.0407	0.0811	0.935479	0.941269	10.369199

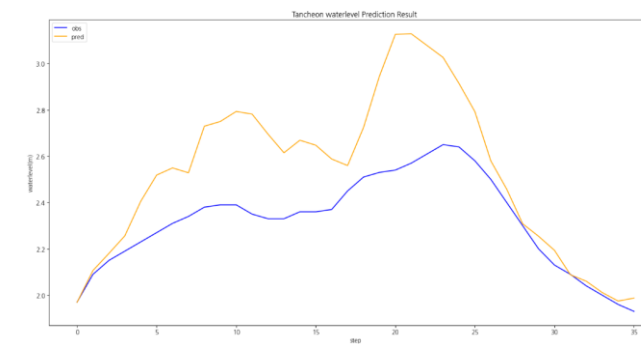
loss_function : mae

예측코드 - 궁내교

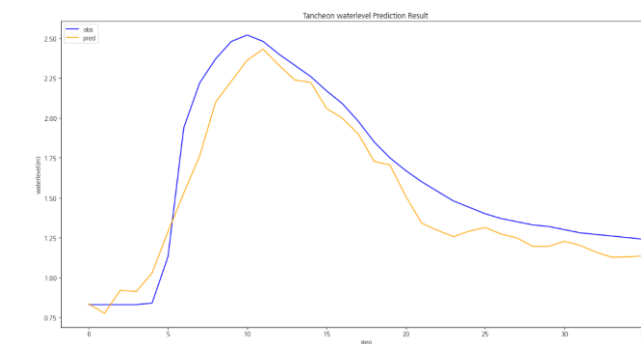
	궁내교
testdata_170723	r2 : 0.9577008281092061 mse : 0.014935925717914087 mae : 0.10249933315647974 rmse : 0.12221262503487144
testdata_220712	r2 : -0.8331744070818663 mse : 0.072737418341244 mae : 0.20990060581101314 rmse : 0.2696987548010632
testdata_230711	r2 : 0.8977044126065418 mse : 0.02862649663022865 mae : 0.13949287262227802 rmse : 0.1691936660464234



testdata_170723



testdata_220712



testdata_230711

IETD

IETD 학습 데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005

홍수통제소 API 호출 → IETD 활용 강우 이벤트 추출 → 관심 강우 포함 파일 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+ 기존 학습 데이터 통합 후 중복 이벤트 제거

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	서울시 (대곡교)_WL	대곡교_Ti	궁내교_Ti
2014-08-20 10:50	0.0	0.0	0.0	0.0	0.0	0.0	0.96	1.84	0.0	0.0

학습 : 총 41 개 中 38개 데이터 사용

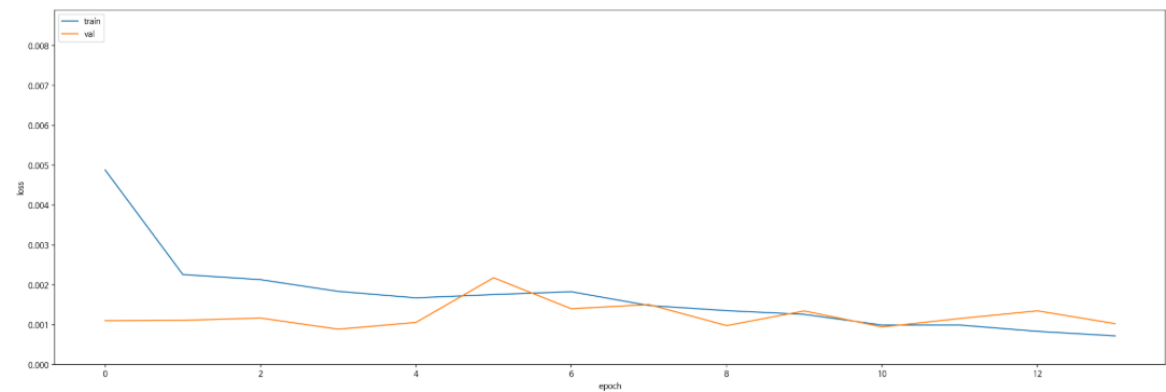
검증 : 학습 데이터에서 제외한 제외한 3개 데이터 (170723, 220713, 230711) 사용

loss_function : mse

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (6325 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0834	0.0510	0.1735	0.897423	0.898354	2.090755

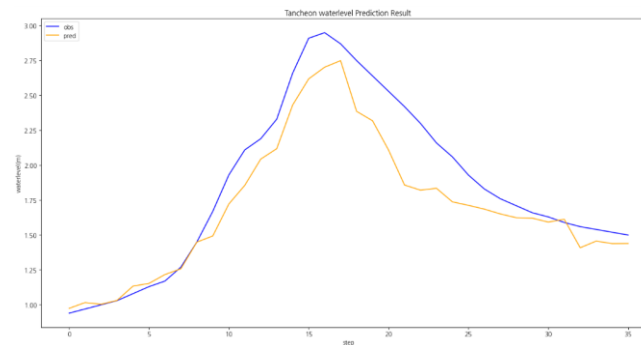
Valid (1439 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1033	0.0881	0.1350	0.882293	0.902797	14.850655

Test (1169 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0994	0.0880	0.1335	0.803966	0.842328	7.905292

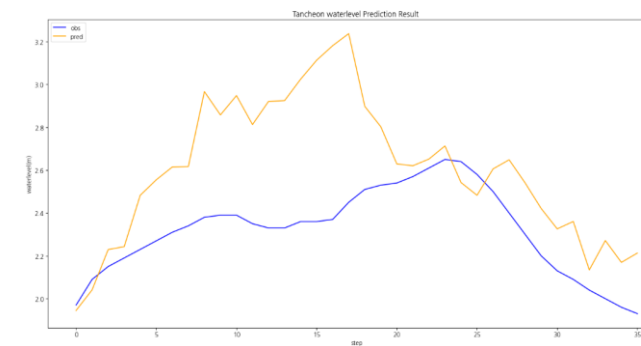
loss_function : mse

예측코드 - 궁내교

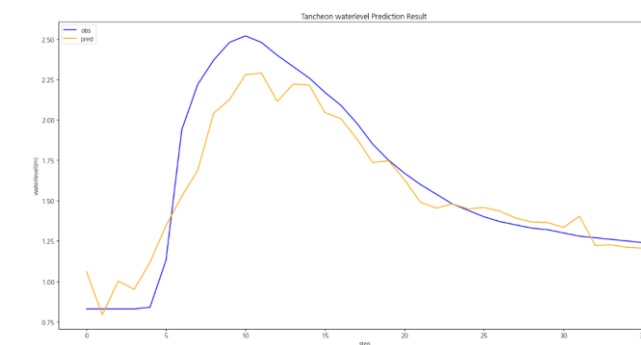
	궁내교
testdata_170723	r2 : 0.8644668641625052 mse : 0.04785703262491908 mae : 0.16468629221121475 rmse : 0.21876250278537016
testdata_220712	r2 : -2.6146979334582174 mse : 0.14342541263256836 mae : 0.30120332029130725 rmse : 0.3787154771494933
testdata_230711	r2 : 0.8808013587825685 mse : 0.0333566636458539 mae : 0.13268791251712372 rmse : 0.1826380673513983



testdata_170723



testdata_220712



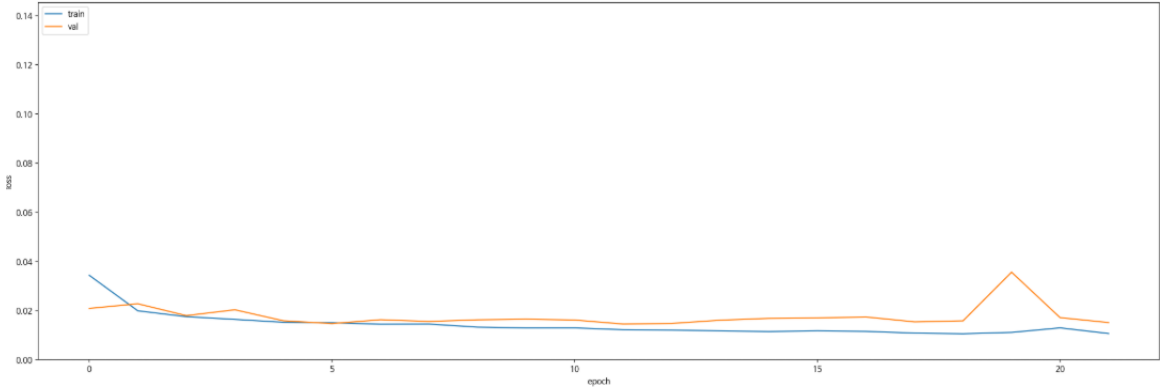
testdata_230711

loss_function : mae

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (6325 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0540	0.0309	0.1639	0.908518	0.911213	0.146762

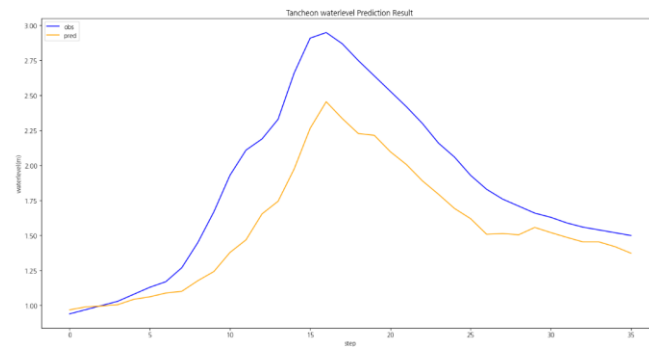
Valid (1439 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0658	0.0510	0.0968	0.939514	0.941580	7.109138

Test (1169 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0648	0.0518	0.0979	0.894547	0.903897	6.465039

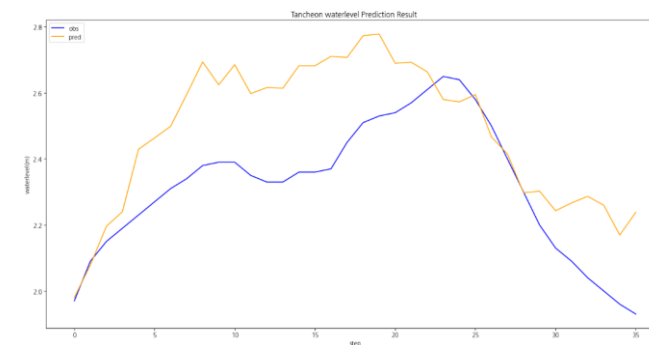
loss_function : mae

예측코드 - 궁내교

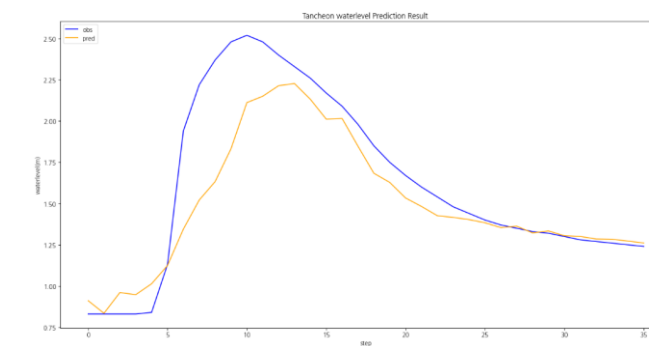
	궁내교
testdata_170723	r2 : 0.6317008228046512 mse : 0.13004720675763015 mae : 0.29296479437086315 rmse : 0.3606205855988121
testdata_220712	r2 : -0.07962368920681295 mse : 0.0428377352583486 mae : 0.17547238654560515 rmse : 0.20697278869056338
testdata_230711	r2 : 0.7661752446992534 mse : 0.06543374685298414 mae : 0.15676571084393393 rmse : 0.25580020886032157



testdata_170723



testdata_220712



testdata_230711

예측에 사용할 데이터 학습에서 제외 2 (증설 전 후 비교)

MIET 학습 데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 강우 포함 파일 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+ 기존 학습 데이터 통합 후 중복 이벤트 제거

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	서울시 (대곡교)_WL	대곡교_Ti	궁내교_Ti
2014-08-20 10:50	0.0	0.0	0.0	0.0	0.0	0.0	0.96	1.84	0.0	0.0

학습 : 총 67개 中 64개 데이터 사용

검증 : 학습 데이터에서 제외한 2개 데이터 (220713, 230711) + 학습에서 검증에 사용된 데이터 (250716, 250814) 사용

궁내교

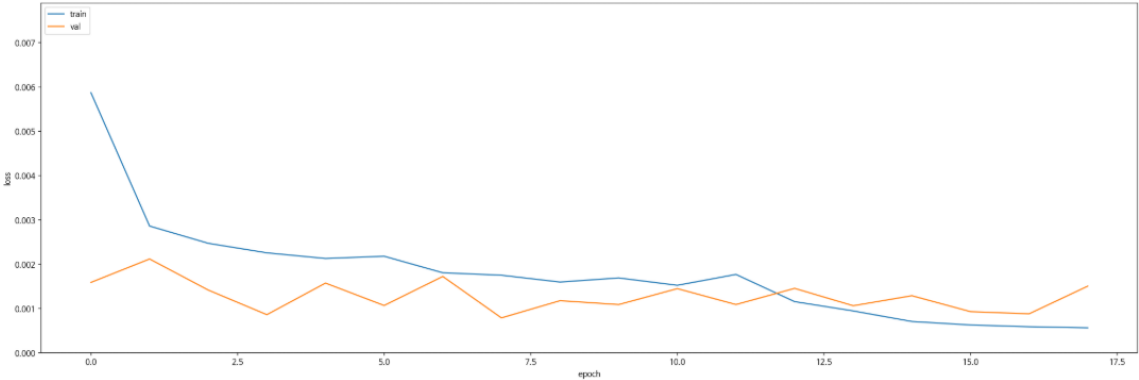
smoothing = 1

loss_function : mse

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0929	0.0566	0.1681	0.916973	0.918180	32.452990

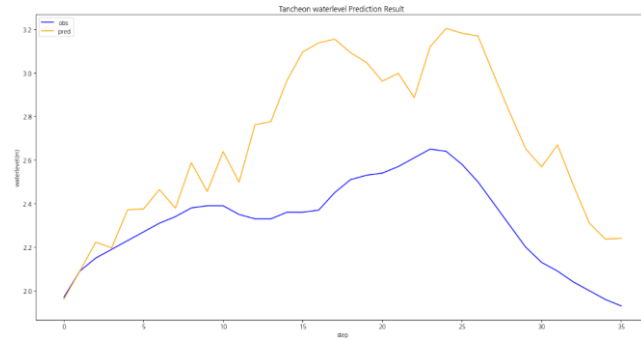
Valid (1247 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0920	0.0681	0.1286	0.893660	0.896028	18.568189

Test (693 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0759	0.0586	0.1097	0.881866	0.890274	4.491322

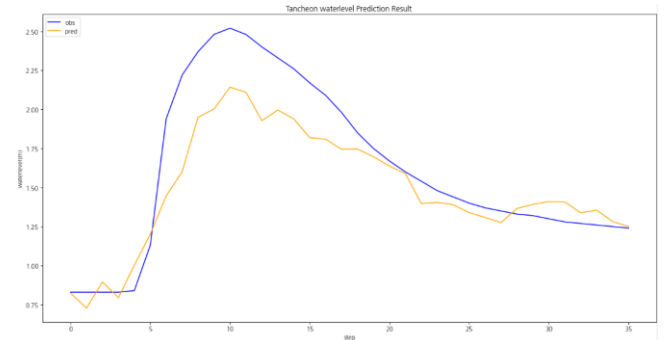
loss_function : mse

예측코드 - 궁내교

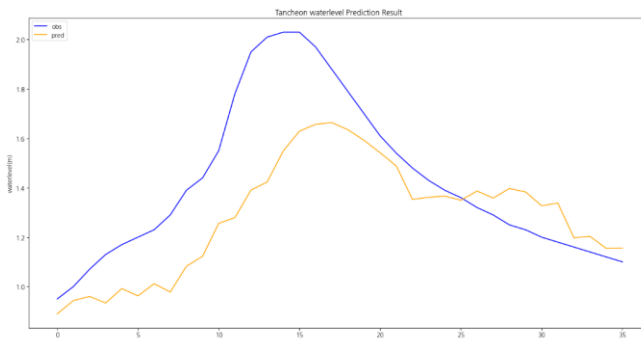
	궁내교
증설 전	testdata_220712 r2 : -3.7460557062787814 mse : 0.18831587329869987 mae : 0.37048792507913375 rmse : 0.43395376861907753
	testdata_230711 r2 : 0.7858148103033311 mse : 0.059937791720316914 mae : 0.17798842834101786 rmse : 0.24482195922816424
증설 후	testdata_250716 r2 : 0.3947407285994061 mse : 0.06072095512751068 mae : 0.19081864602035944 rmse : 0.24641622334479255
	testdata_250814 r2 : -0.056341420985954516 mse : 0.07447427089080348 mae : 0.2094912936952379 rmse : 0.27289974512777304



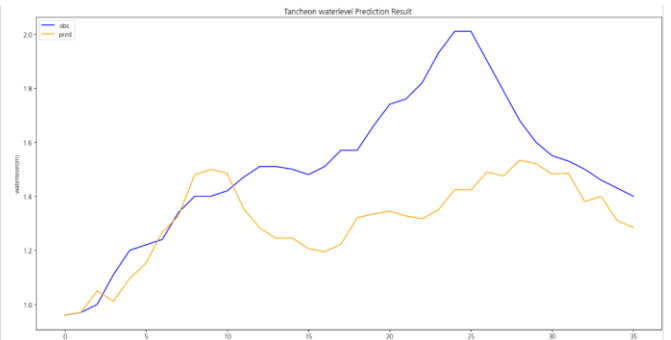
testdata_220712



testdata_230711



testdata_250716



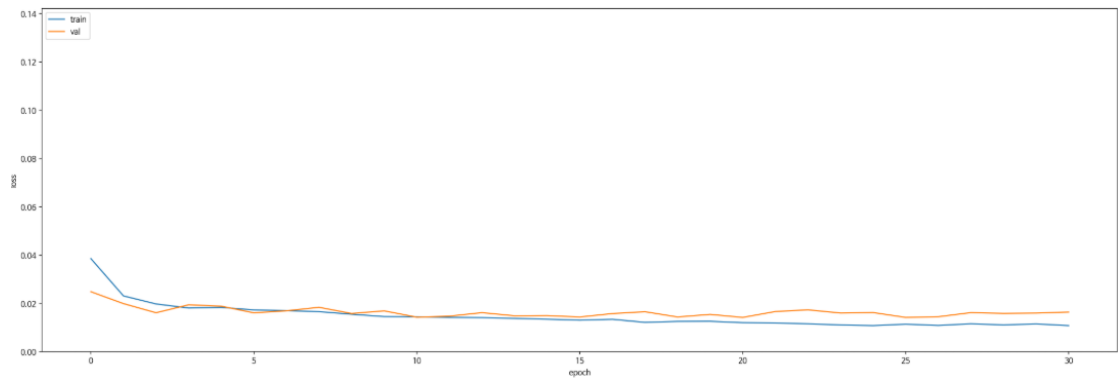
testdata_250814

loss_function : mae

학습 코드 - 궁내교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0567	0.0316	0.1677	0.917413	0.921557	5.921931

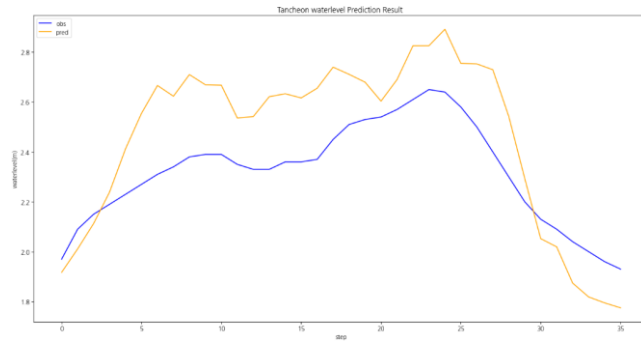
Valid (1247 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0651	0.0469	0.0974	0.939098	0.939761	8.015105

Test (693 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0538	0.0395	0.0869	0.925799	0.929416	6.737624

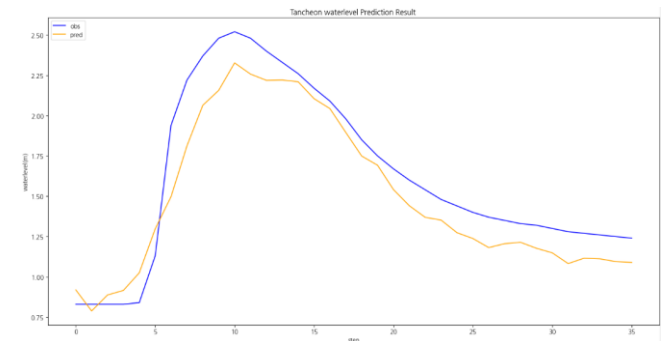
loss_function : mae

예측코드 - 궁내교

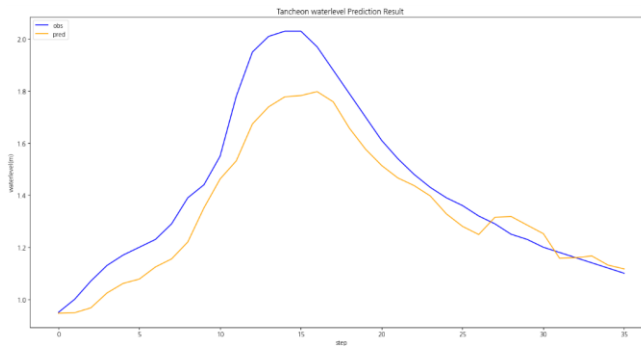
	궁내교
증설 전	testdata_220712 r2 : -0.17952902064759457 mse : 0.046801818468028854 mae : 0.19706654959254793 rmse : 0.21633727942273115
	testdata_230711 r2 : 0.8818314085091158 mse : 0.03306841352893966 mae : 0.15780521591504415 rmse : 0.18184722579390553
증설 후	testdata_250716 r2 : 0.8383801421544474 mse : 0.016214063294305487 mae : 0.10172216607464683 rmse : 0.12733445446659553
	testdata_250814 r2 : 0.5624161614966373 mse : 0.030850572247483943 mae : 0.12672317968474495 rmse : 0.17564330971455744



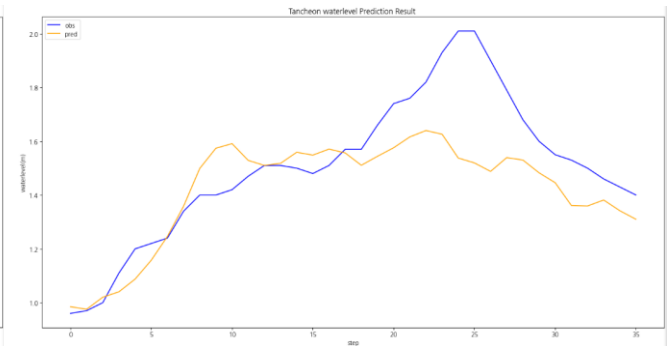
testdata_220712



testdata_230711



testdata_250716



testdata_250814

대곡교

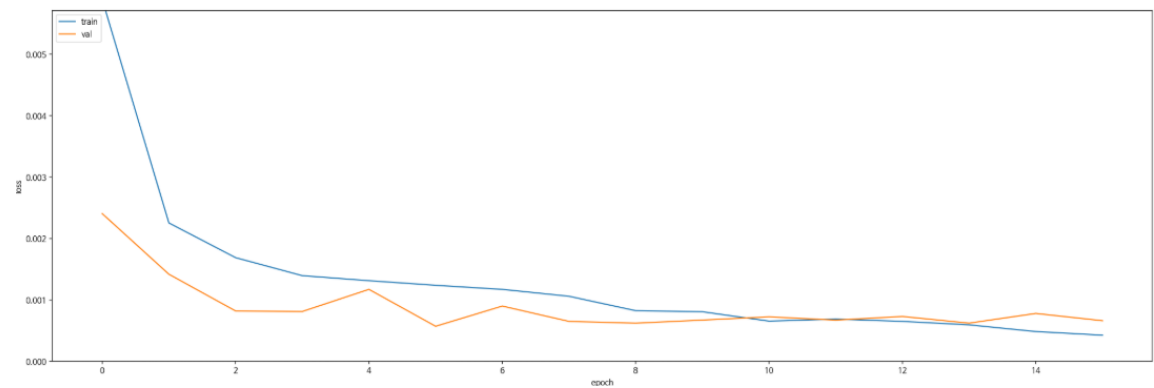
smoothing = 2

loss_function : mse

학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1264	0.0467	0.2115	0.940239	0.943470	9.337945

Valid (1247 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1118	0.0427	0.1532	0.937498	0.939542	1.007802

Test (693 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.1024	0.0390	0.1519	0.912317	0.914719	11.241669

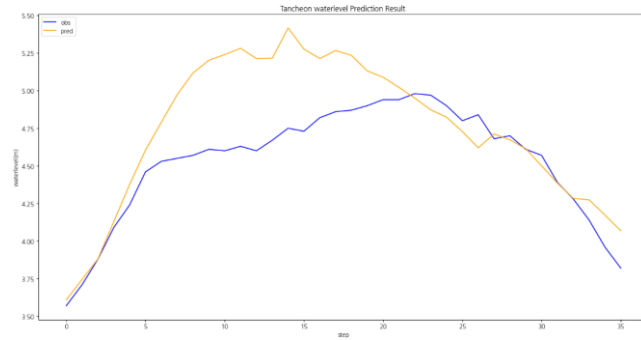
loss_function : mse

예측코드 - 대곡교

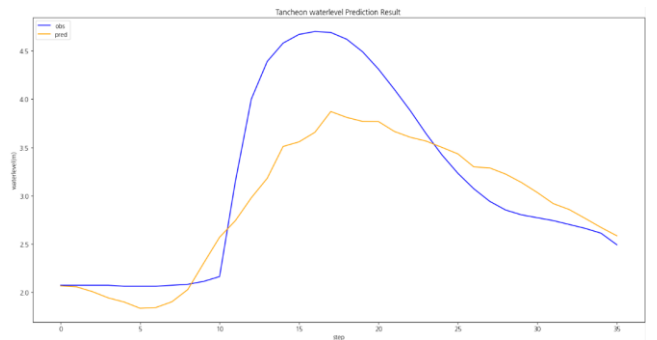
대곡교	
testdata_220712	r2 : 0.22127644426133286 mse : 0.10905206219080067 mae : 0.24274081362618333 rmse : 0.3302303168862615
testdata_230711	r2 : 0.7078427228575586 mse : 0.26590938043516943 mae : 0.3782702791690826 rmse : 0.5156640189456401
testdata_250716	r2 : 0.9009047332623752 mse : 0.04576081785628601 mae : 0.18652880761358479 rmse : 0.21391778293607572
testdata_250814	r2 : 0.6478002839292782 mse : 0.12224816816139523 mae : 0.28486741065979004 rmse : 0.3496400551444231

증설 전

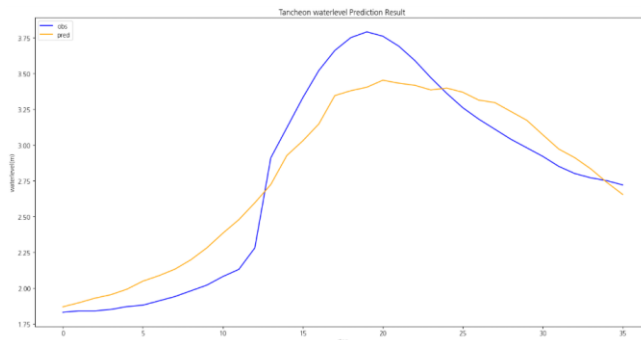
증설 후



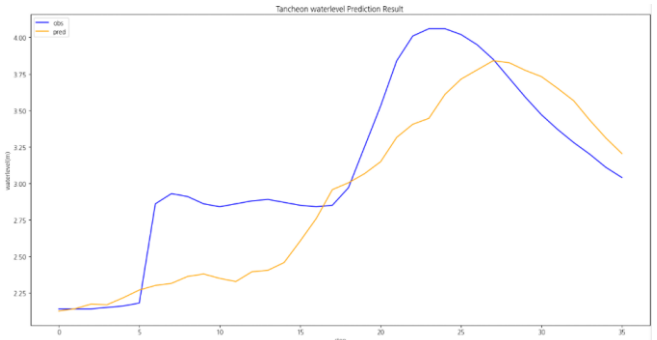
testdata_220712



testdata_230711



testdata_250716



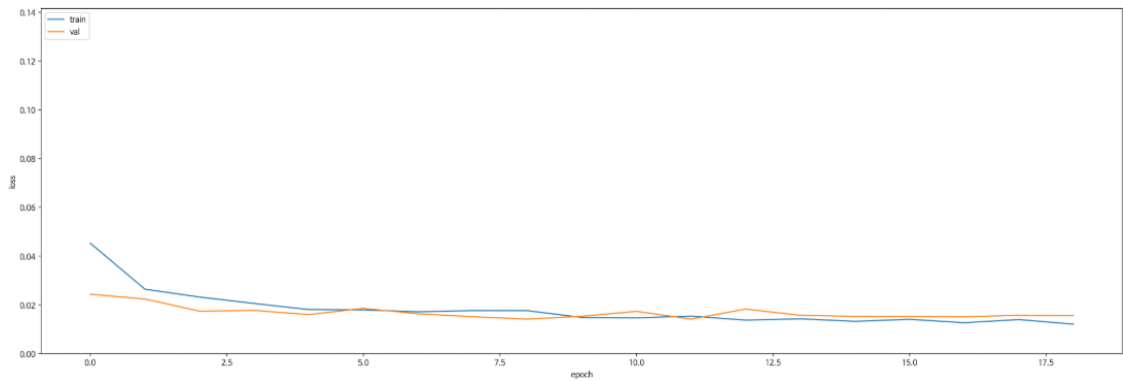
testdata_250814

loss_function : mae

학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5199 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0955	0.0341	0.1955	0.948953	0.954236	7.418984

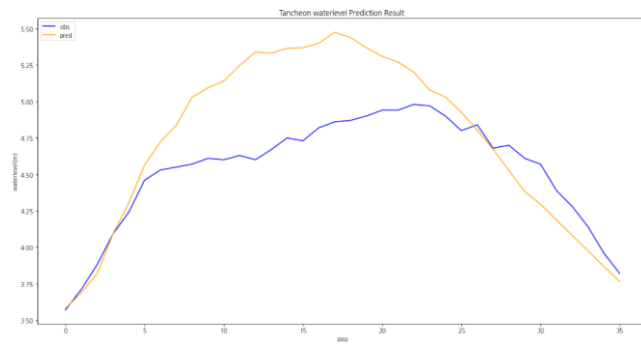
Valid (1247 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0907	0.0332	0.1312	0.954134	0.954752	2.454251

Test (693 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0863	0.0313	0.1373	0.928315	0.932271	7.094862

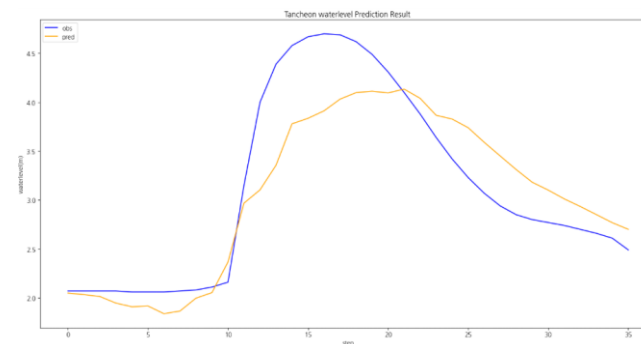
loss_function : mae

예측코드 - 대곡교

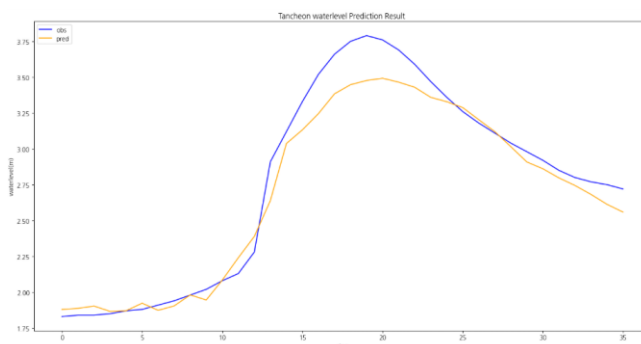
	대곡교
증설 전	testdata_220712 r2 : 0.029128645059933733 mse : 0.13596034510316254 mae : 0.2899722901980082 rmse : 0.3687280096536776
	testdata_230711 r2 : 0.7954479485971804 mse : 0.18617475418470464 mae : 0.33957783407635156 rmse : 0.4314797262730946
증설 후	testdata_250716 r2 : 0.955878028785127 mse : 0.020374913501872607 mae : 0.10596535987324186 rmse : 0.14274072124615528
	testdata_250814 r2 : 0.8412813204617778 mse : 0.05509109445916487 mae : 0.18988143099678878 rmse : 0.23471492167982178



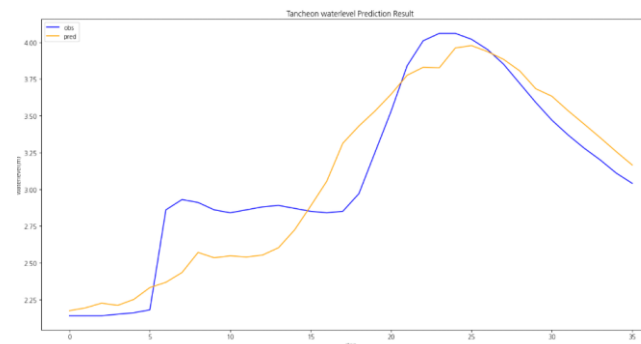
testdata_220712



testdata_230711



testdata_250716



testdata_250814

* model_(gn/dg)_(mse/mae)_2

* model_(gn/dg)_(mse/mae)_3

* model_(gn/dg)_(mse/mae)_4

* model_(gn/dg)_(mse/mae)_5

예측에 사용할 데이터 학습에서
제외 3 , learning rate 변경,
학습데이터 순서 변경
(증설 전 후 비교)

MIET 학습 데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005 , 0.0001 , 0.0003 , 0.00005

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 강우 포함 파일 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+ 기존 학습 데이터 통합 후 중복 이벤트 제거

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	서울시 (대곡교)_WL	대곡교_Ti	궁내교_Ti
2014-08-20 10:50	0.0	0.0	0.0	0.0	0.0	0.0	0.96	1.84	0.0	0.0

학습 : 총 67개 中 64개 데이터 사용 (2014 년 강우이벤트를 Train에서 Test로 이동)

검증 : 학습 데이터에서 제외한 2개 데이터 (220713, 230711) + 학습에서 검증에 사용된 데이터 (250716, 250814) 사용

* model_(gn/dg)_(mse/mae)2

learning_rate = 0.0005

learning_rate = 0.0001

궁내교

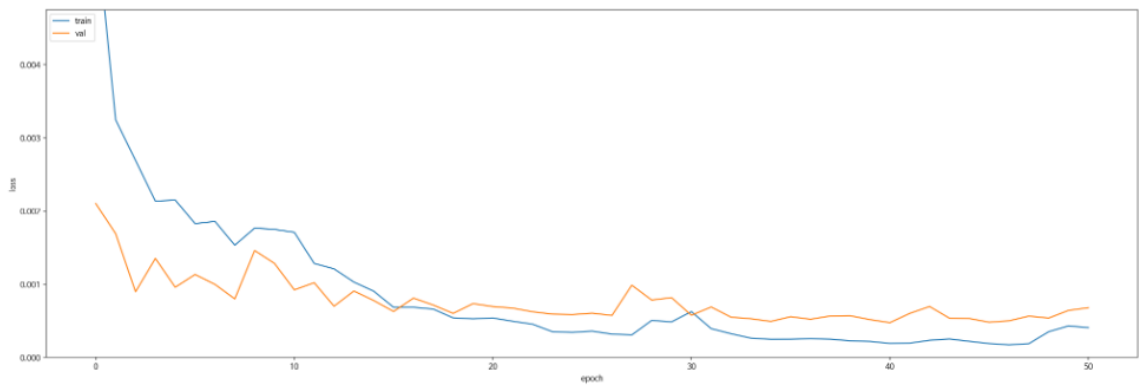
smoothing = 1

loss_function : mse

학습 코드 - 궁내교

Train : Valid : Test = 5 : 1 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5112 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0406	0.0264	0.0632	0.988320	0.988722	8.370593

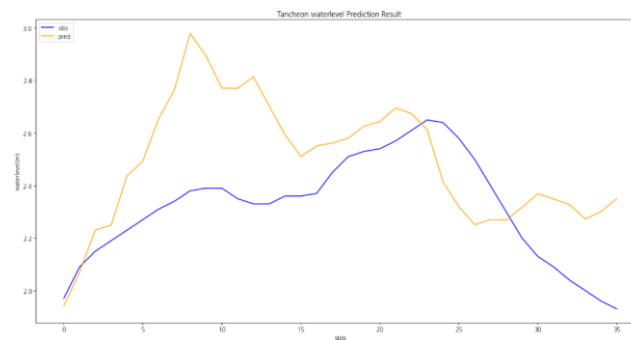
Valid (1118 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0738	0.0590	0.0997	0.915736	0.924870	6.234111

Test (909 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0566	0.0392	0.0930	0.930270	0.932822	0.277249

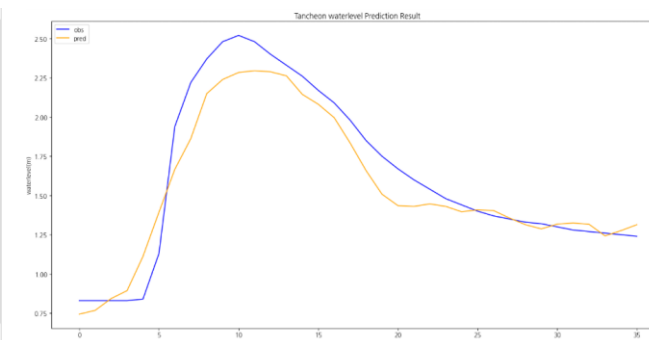
loss_function : mse

예측코드 - 궁내교

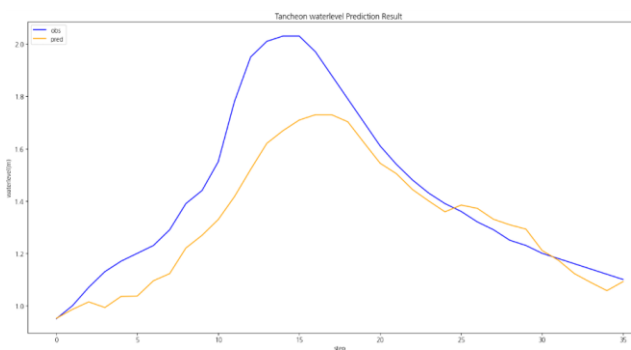
궁내교	
증설 전	testdata_220712 r2 : -0.8660661448633538 mse : 0.07404250970720465 mae : 0.22651568240589567 rmse : 0.2721075333525418
증설 후	testdata_230711 r2 : 0.9173872788953554 mse : 0.023118424191847026 mae : 0.11795920928319292 rmse : 0.1520474405961739
증설 후	testdata_250716 r2 : 0.7104807689773868 mse : 0.029045212632257497 mae : 0.12238042977121137 rmse : 0.17042656081801774
증설 후	testdata_250814 r2 : 0.85113363889279 mse : 0.010495388596310606 mae : 0.08006758908430733 rmse : 0.10244700384252634



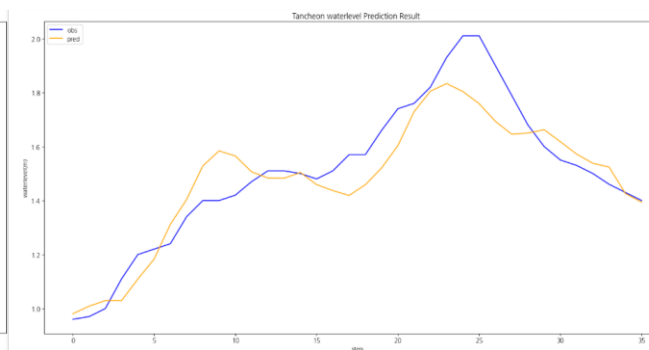
testdata_220712



testdata_230711



testdata_250716



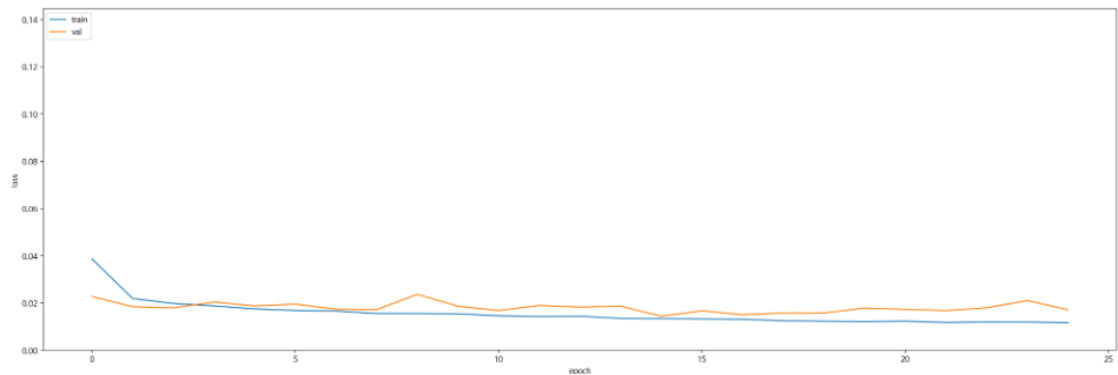
testdata_250814

loss_function : mae

학습 코드 - 궁내교

Train : Valid : Test = 5 : 1 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5112 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0610	0.0333	0.1854	0.899597	0.903558	1.632025

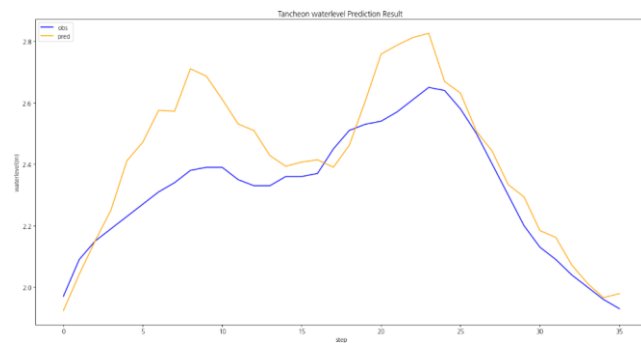
Valid (1118 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0661	0.0511	0.1002	0.914883	0.920608	10.540136

Test (909 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0531	0.0353	0.0967	0.924536	0.930873	5.521413

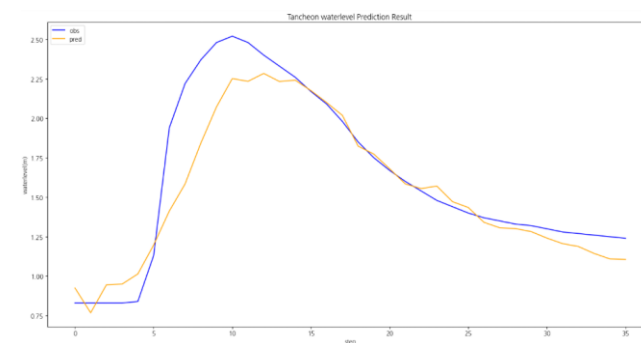
loss_function : mae

예측코드 - 궁내교

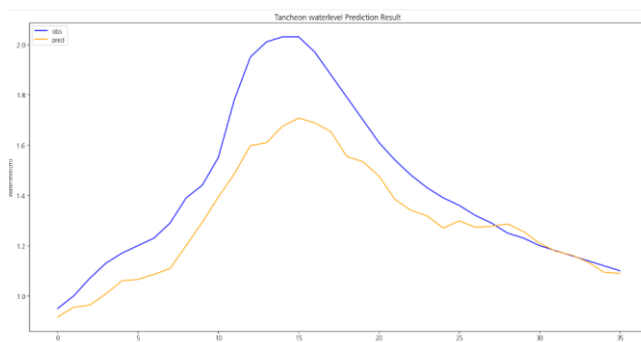
	궁내교
증설 전	testdata_220712 r2 : 0.48350324625474395 mse : 0.020493762243204518 mae : 0.10971201062202456 rmse : 0.1431564257838415
	testdata_230711 r2 : 0.8566634699470248 mse : 0.04011143392498437 mae : 0.12565590659777326 rmse : 0.20027839105850728
증설 후	testdata_250716 r2 : 0.6931006243265065 mse : 0.03078882736617747 mae : 0.13623218384053967 rmse : 0.17546745386588783
	testdata_250814 r2 : 0.6177622192625288 mse : 0.02694855987070158 mae : 0.1293952695528666 rmse : 0.1641601652981063



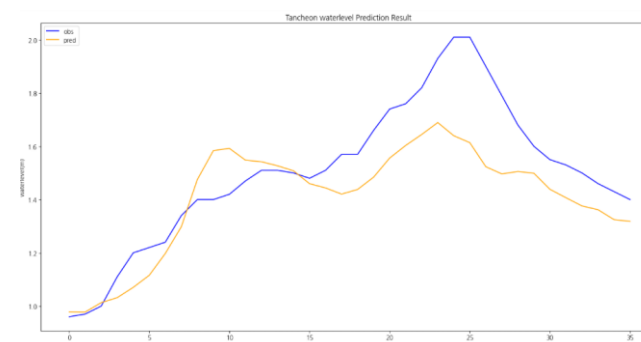
testdata_220712



testdata_230711



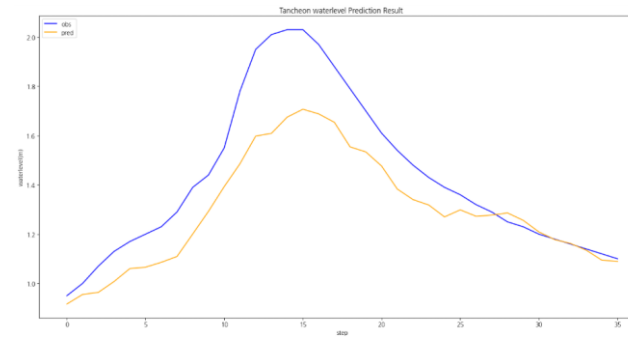
testdata_250716



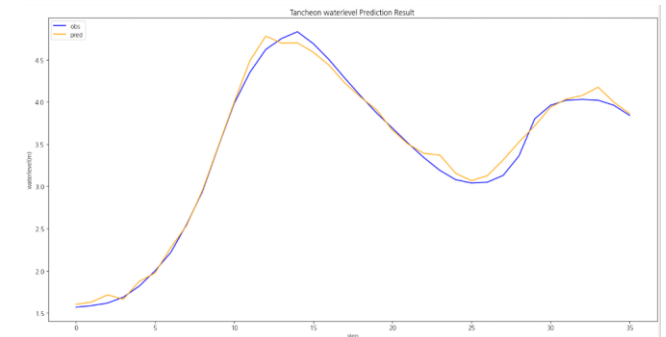
testdata_250814

+) 예측코드 - 궁내교 (학습에도 사용 된 데이터)

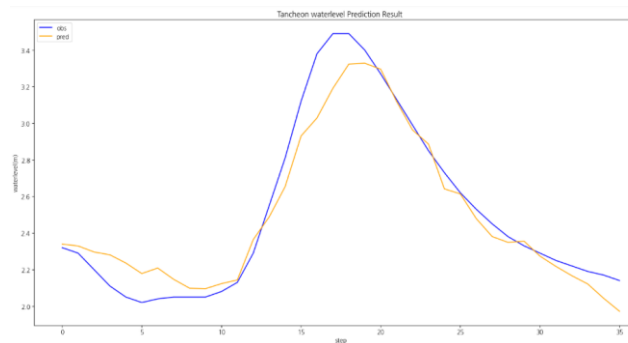
	궁내교
mse	testdata_220630 r2 : 0.6931006243265065 mse : 0.03078882736617747 mae : 0.13623218384053967 rmse : 0.17546745386588783
	testdata_220808 r2 : 0.9924956911191448 mse : 0.006907648577594287 mae : 0.06417400201161705 rmse : 0.08311226490473164
mae	testdata_220630 r2 : 0.9328815422847411 mse : 0.014795373233673977 mae : 0.09148696263631188 rmse : 0.12163623322708565
	testdata_220808 r2 : 0.9818703163715436 mse : 0.01668821010923914 mae : 0.10055554813808862 rmse : 0.1291828553223652



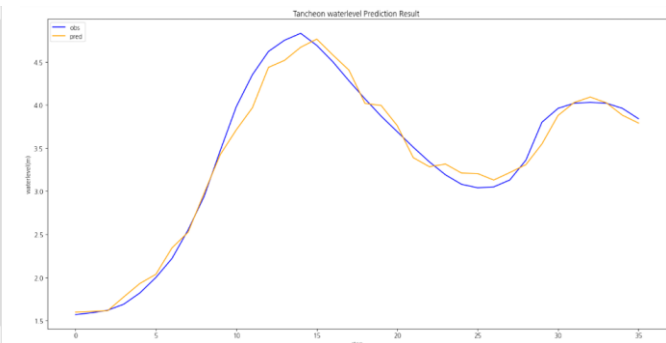
testdata_220630
(mse)



testdata_220808
(mse)



testdata_220630
(mae)



testdata_220808
(mae)

대곡교

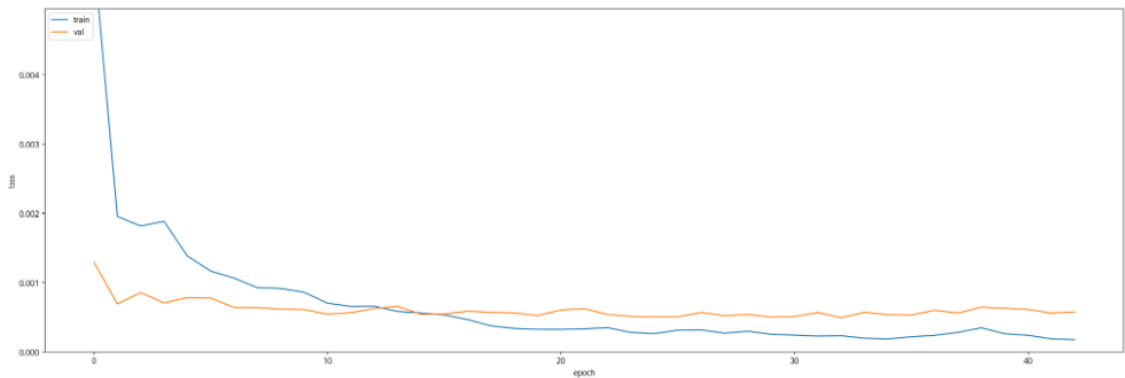
smoothing = 2

loss_function : mse

학습 코드 - 대곡교

Train : Valid : Test = 5 : 1 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5112 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0532	0.0188	0.0845	0.990552	0.990683	3.483236

Valid (1118 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0995	0.0384	0.1426	0.927635	0.927990	4.866550

Test (909 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0775	0.0289	0.1231	0.953586	0.953834	10.480178

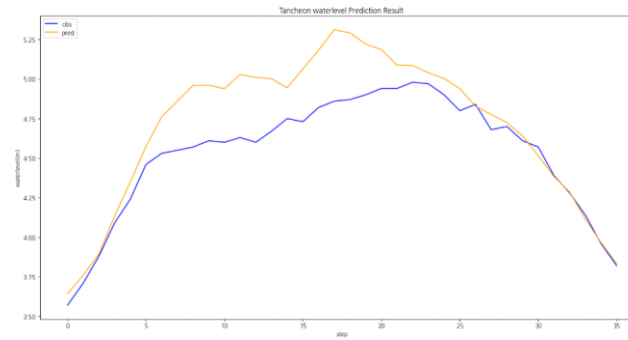
loss_function : mse

예측코드 - 대곡교

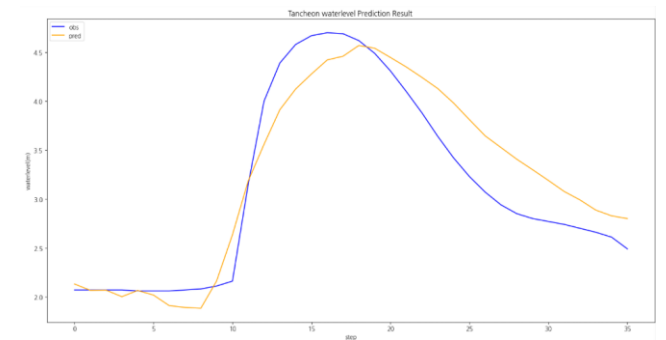
대곡교	
testdata_220712	r2 : 0.6192848002881184 mse : 0.053315168560145905 mae : 0.17582903226216626 rmse : 0.23090077643902782
testdata_230711	r2 : 0.8730247156992079 mse : 0.11556761313373512 mae : 0.2791214652856191 rmse : 0.3399523689191401
증설 후	
testdata_250716	r2 : 0.9576316738162141 mse : 0.019565104582697595 mae : 0.11279573056432939 rmse : 0.13987531798962102
testdata_250814	r2 : 0.8403428495808809 mse : 0.05541683676055683 mae : 0.1667659605873955 rmse : 0.2354078094723215

증설 전

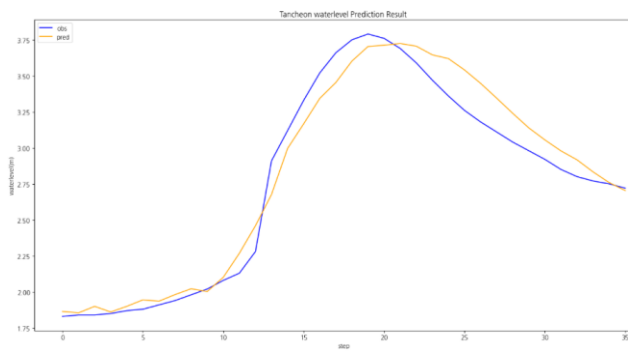
증설 후



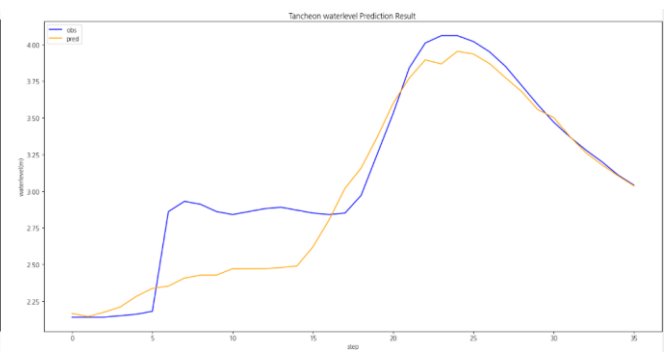
testdata_220712



testdata_230711



testdata_250716



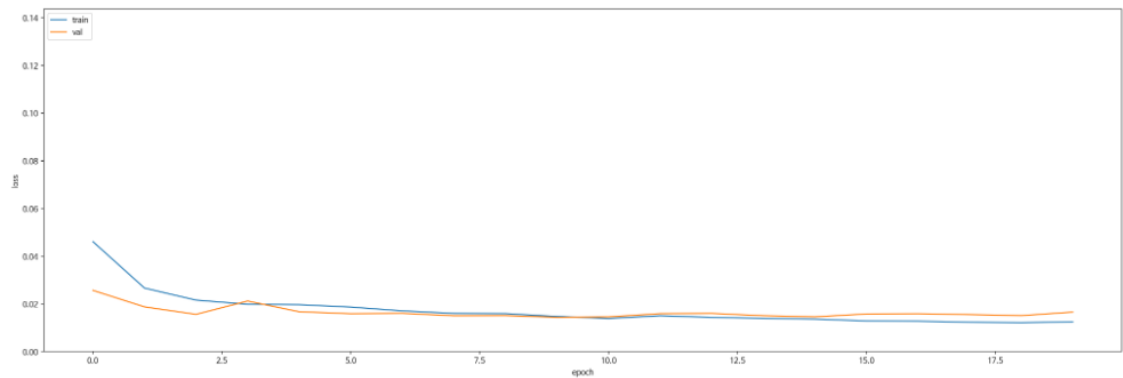
testdata_250814

loss_function : mae

학습 코드 - 대곡교

Train : Valid : Test = 5 : 1 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (5112 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0917	0.0329	0.1823	0.955965	0.958530	4.670675

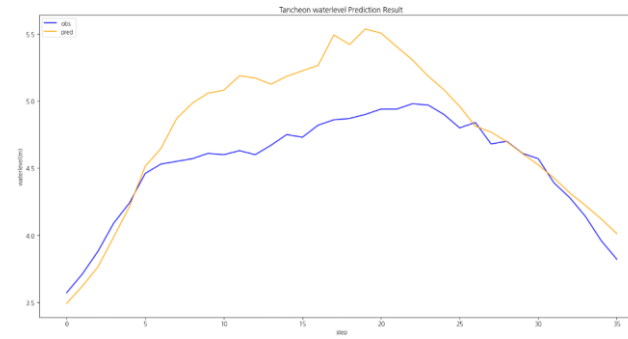
Valid (1118 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0920	0.0353	0.1294	0.940395	0.944118	5.373765

Test (909 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0834	0.0316	0.1313	0.947177	0.951074	10.481600

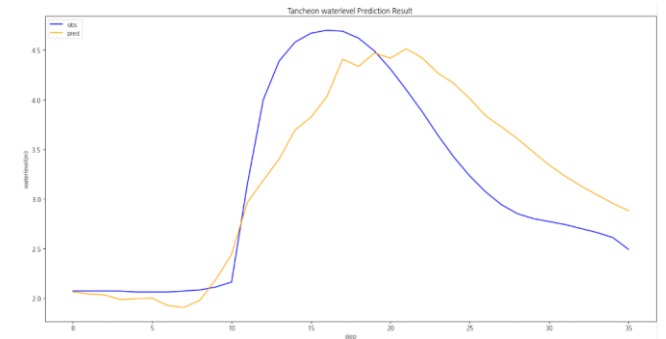
loss_function : mae

예측코드 - 대곡교

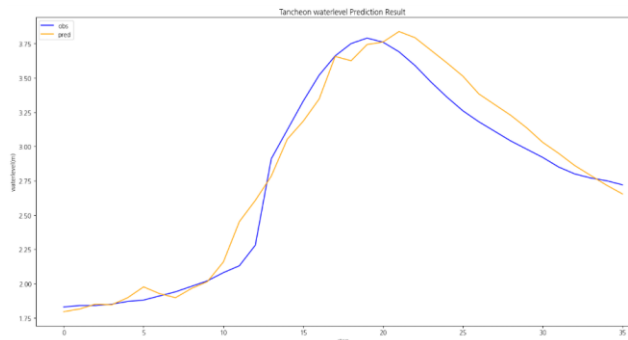
	대곡교
증설 전	testdata_220712 r2 : 0.1707880218587715 mse : 0.11612243593150111 mae : 0.267141786151462 rmse : 0.34076742205131805
	testdata_230711 r2 : 0.714071445867047 mse : 0.26024025628205844 mae : 0.4118883290555742 rmse : 0.5101374876266774
증설 후	testdata_250716 r2 : 0.9559205520047541 mse : 0.020355276869649035 mae : 0.10858701308568325 rmse : 0.1426719203965834
	testdata_250814 r2 : 0.8317512041579551 mse : 0.0583989882686915 mae : 0.20596396313773252 rmse : 0.24165882617585374



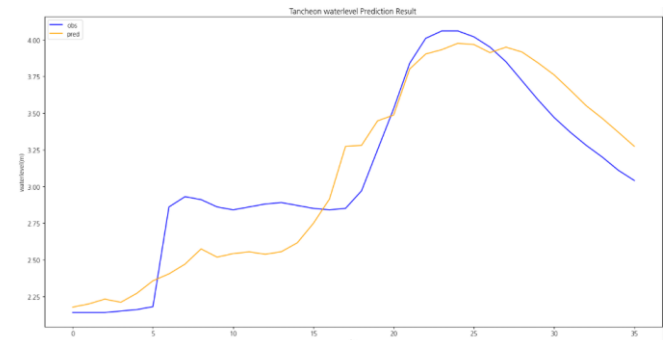
testdata_220712



testdata_230711



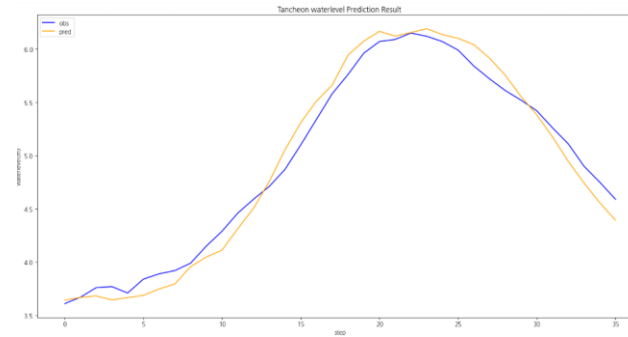
testdata_250716



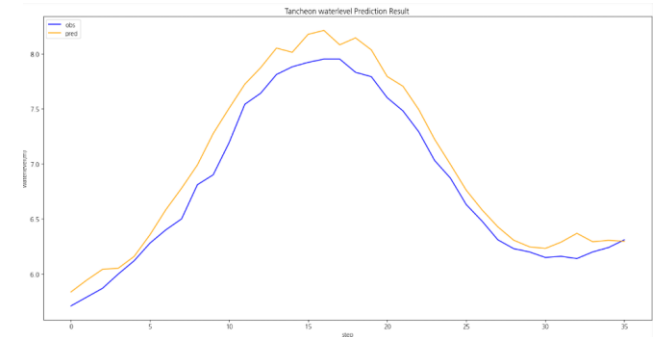
testdata_250814

+) 예측코드 - 대곡교 (학습에도 사용 된 데이터)

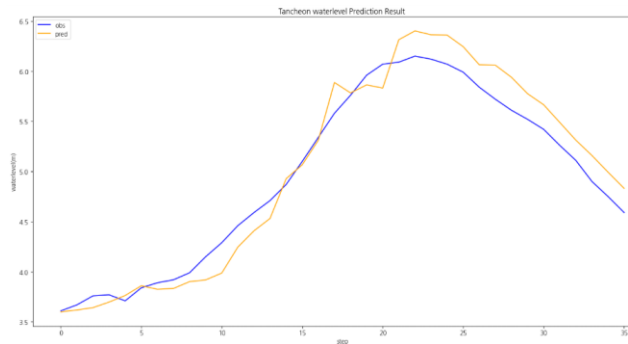
	대곡교
mse	testdata_220630 r2 : 0.9777236746934758 mse : 0.016233493919632053 mae : 0.11147989061143665 rmse : 0.12741072921709556
	testdata_220808 r2 : 0.9339835003971326 mse : 0.034606867864965064 mae : 0.16529195626576734 rmse : 0.18602921239677672
mae	testdata_220630 r2 : 0.9442066711562173 mse : 0.04065844128592935 mae : 0.17462185700734456 rmse : 0.2016393842629196
	testdata_220808 r2 : 0.8683925832391469 mse : 0.0689906388446929 mae : 0.2161240265104505 rmse : 0.2626606914722736



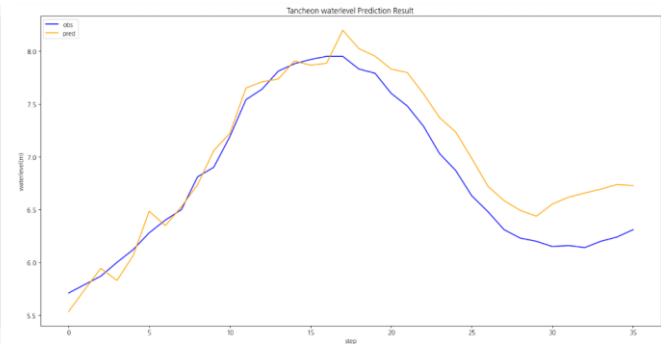
testdata_220630
(mse)



testdata_220808
(mse)



testdata_220630
(mae)



testdata_220808
(mae)

learning_rate = 0.0003
= 0.0001 과 동일

learning_rate = 0.00005
= 0.0001 과 동일

* model_(gn/dg)_(mse/mae)_3

학습 데이터 증강 &
예측 데이터 학습에서 제외 &
학습률 0.0005 -> 0.0001 조정

발표 자료

MIET 학습 데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- learning_rate = 0.0005 → 0.0001

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 강우 포함 파일 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+ 기존 학습 데이터 통합 후 중복 이벤트 제거

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	서울시 (대곡교)_WL	대곡교_Ti	궁내교_Ti
2014-08-20 10:50	0.0	0.0	0.0	0.0	0.0	0.0	0.96	1.84	0.0	0.0

학습 : 총 67개 中 64개 데이터 사용 (2014 년 강우이벤트를 Train에서 Test로 이동)

검증 : 학습 데이터에서 제외한 2개 데이터 (220713, 230711) + 학습에서 검증에 사용된 데이터 (250716, 250814) 사용

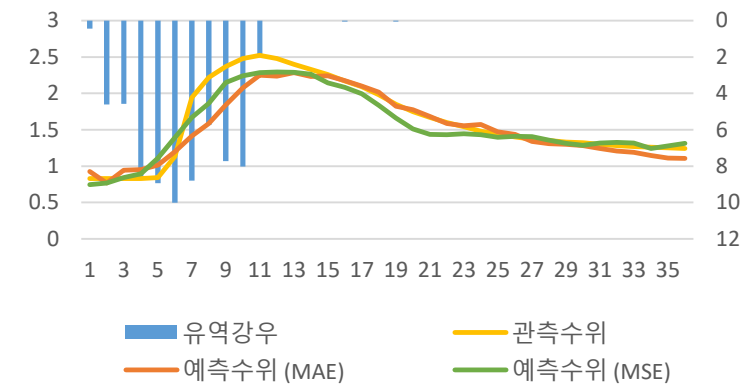
궁내교

smoothing = 1

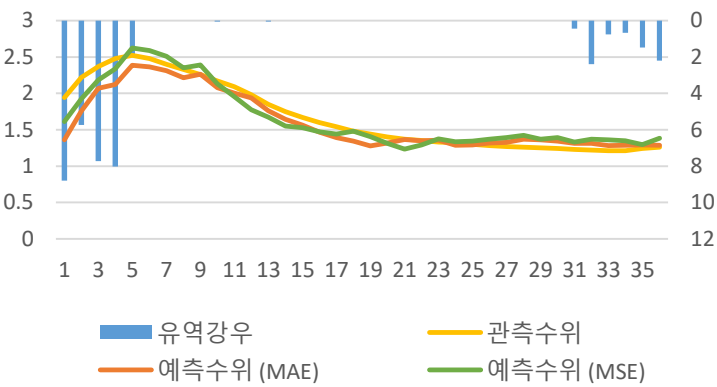
궁내교

20230711		MSE	MAE
07시	R2	0.92	0.86
	MSE	0.02	0.04
	MAE	0.12	0.13
	RMSE	0.15	0.20
08시	R2	0.91	0.86
	MSE	0.02	0.03
	MAE	0.12	0.12
	RMSE	0.14	0.17
09시	R2	0.65	0.78
	MSE	0.04	0.03
	MAE	0.16	0.14
	RMSE	0.21	0.16
10시	R2	-0.26	0.11
	MSE	0.03	0.02
	MAE	0.16	0.13
	RMSE	0.17	0.14

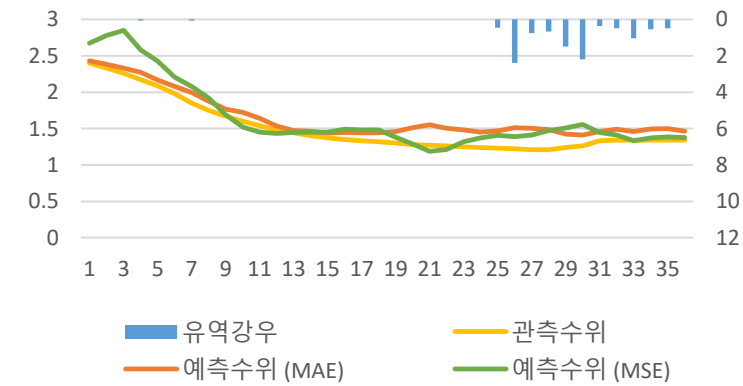
20230711 07시 궁내교



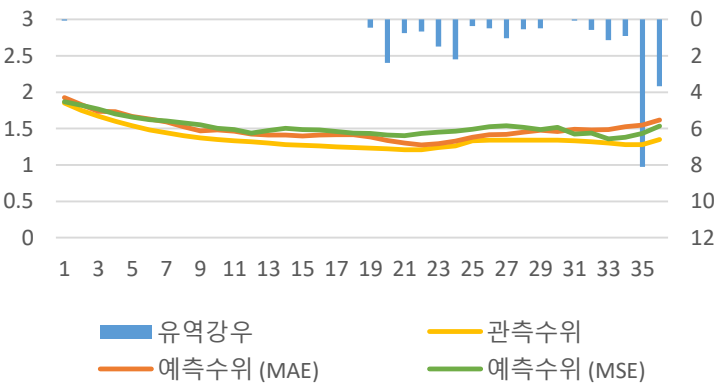
20230711 08시 궁내교



20230711 09시 궁내교

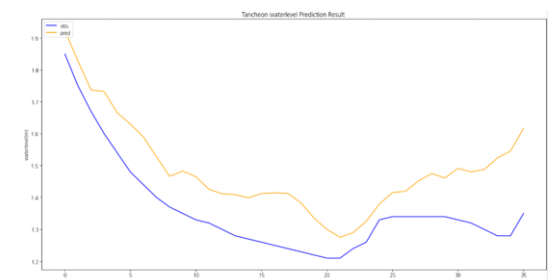
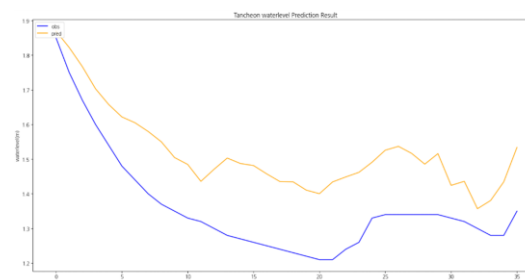
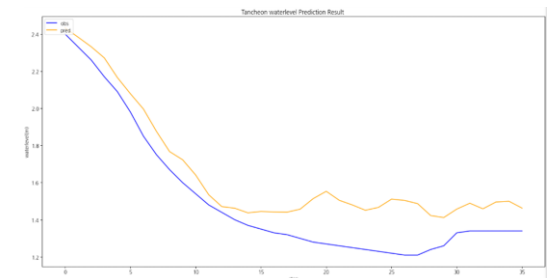
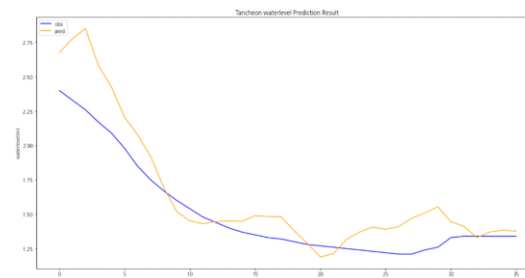
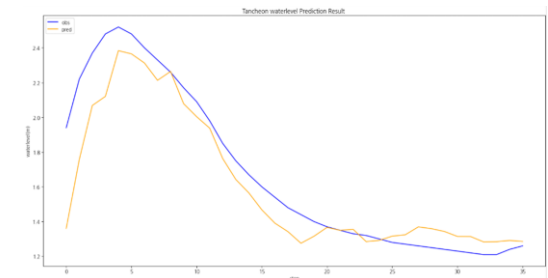
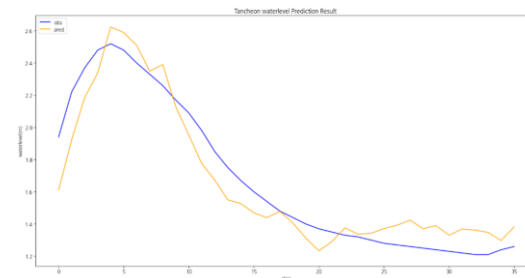
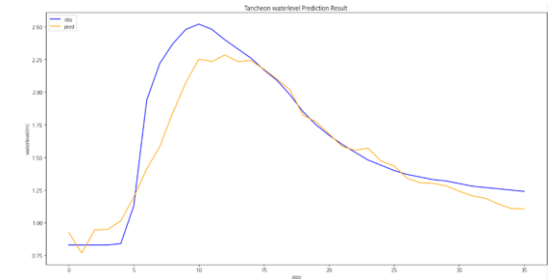
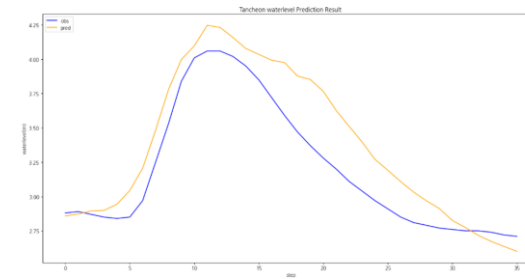


20230711 10시 궁내교



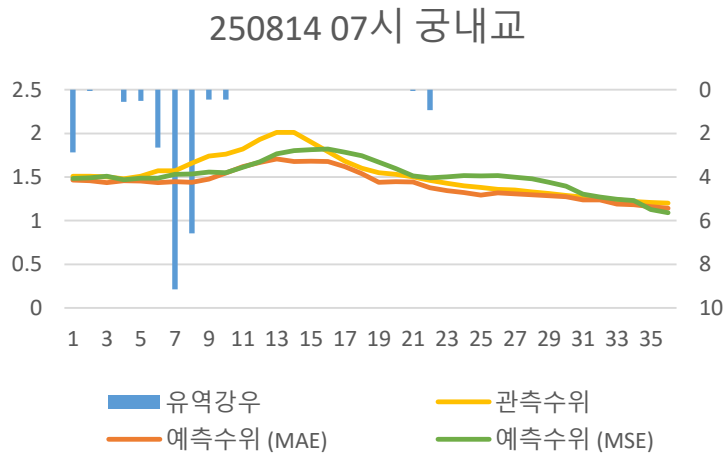
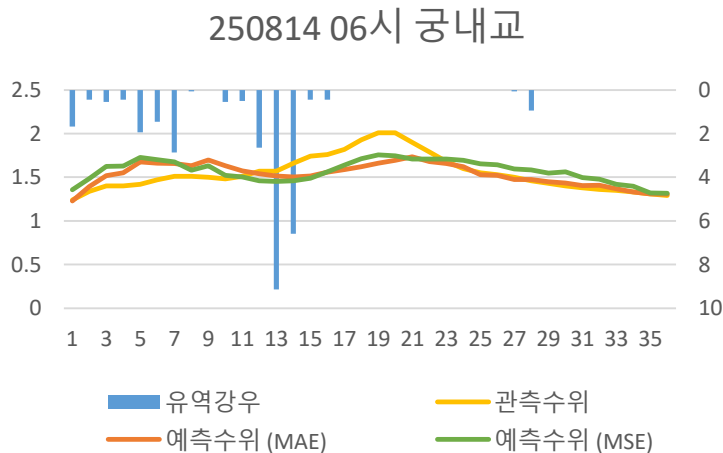
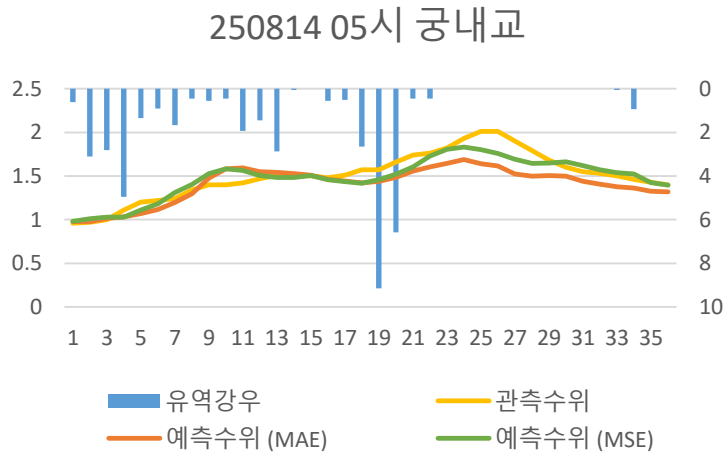
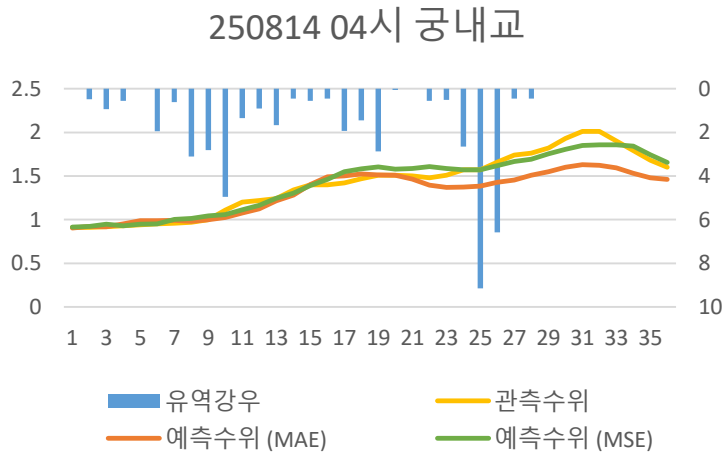
* R2(%) MSE(m²) MAE(m) RMSE(m)

궁내교



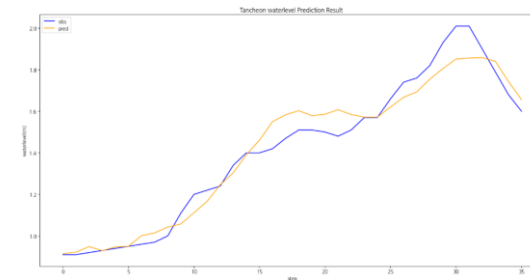
궁내교

20250814		MSE	MAE
04시	R2	0.96	0.75
	MSE	0.01	0.03
	MAE	0.06	0.12
	RMSE	0.07	0.17
05시	R2	0.84	0.62
	MSE	0.06	0.03
	MAE	0.17	0.13
	RMSE	0.24	0.16
06시	R2	0.39	0.47
	MSE	0.02	0.02
	MAE	0.14	0.11
	RMSE	0.16	0.15
07시	R2	0.72	0.65
	MSE	0.01	0.02
	MAE	0.10	0.10
	RMSE	0.12	0.14

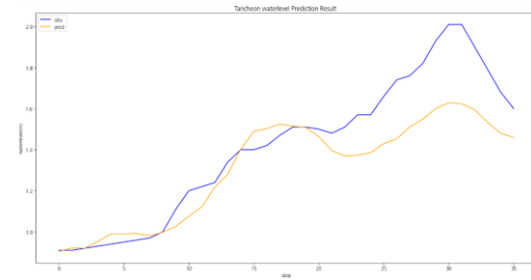


* R2(%) MSE(m²) MAE(m) RMSE(m)

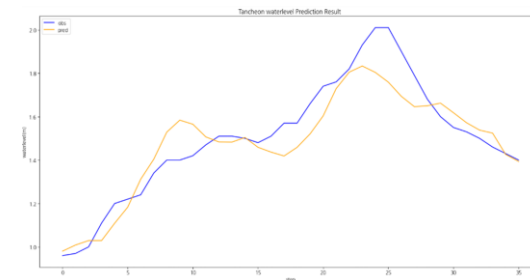
궁내교



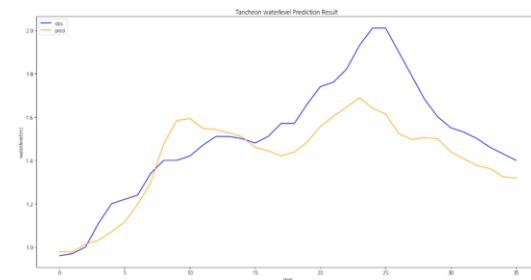
04_mse



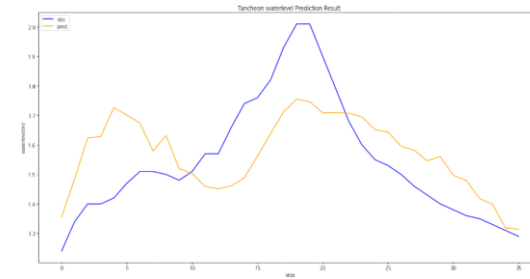
04_mae



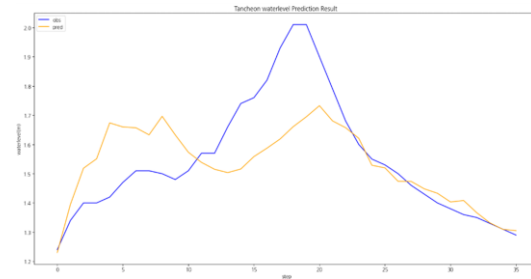
05_mse



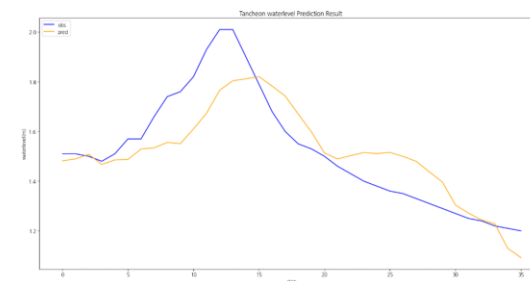
05_mae



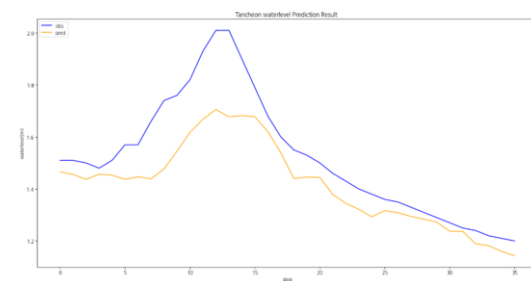
06_mse



06_mae



07_mse



07_mae

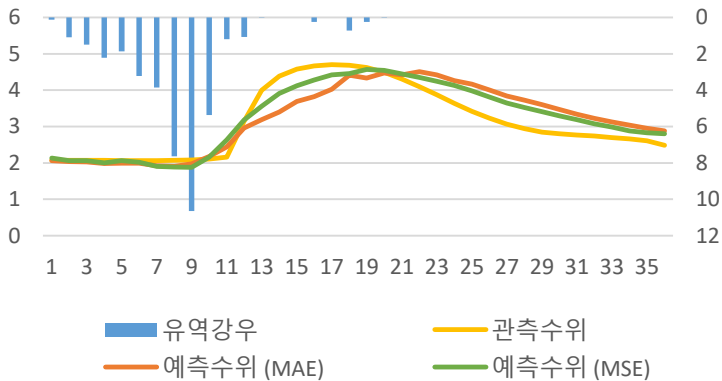
대곡교

smoothing = 2

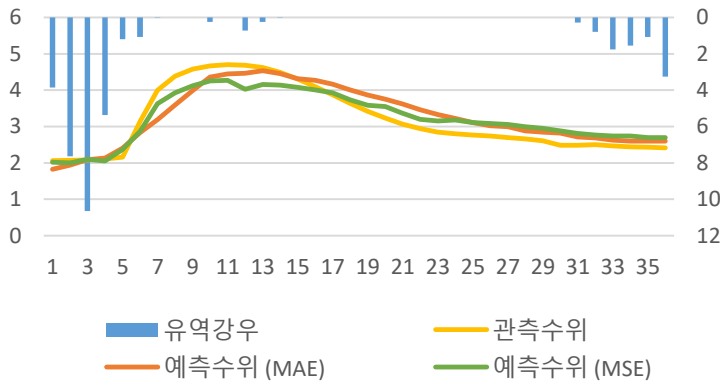
대곡교

20230711		MSE	MAE
07시	R2	0.87	0.71
	MSE	0.12	0.26
	MAE	0.28	0.41
	RMSE	0.34	0.51
08시	R2	0.87	0.84
	MSE	0.10	0.12
	MAE	0.29	0.30
	RMSE	0.32	0.35
09시	R2	0.95	0.85
	MSE	0.03	0.10
	MAE	0.16	0.25
	RMSE	0.19	0.32
10시	R2	0.91	0.84
	MSE	0.06	0.10
	MAE	0.21	0.25
	RMSE	0.24	0.32

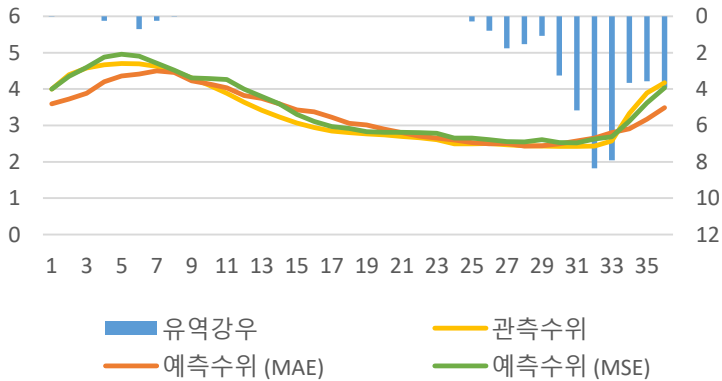
20230711 07시 대곡교



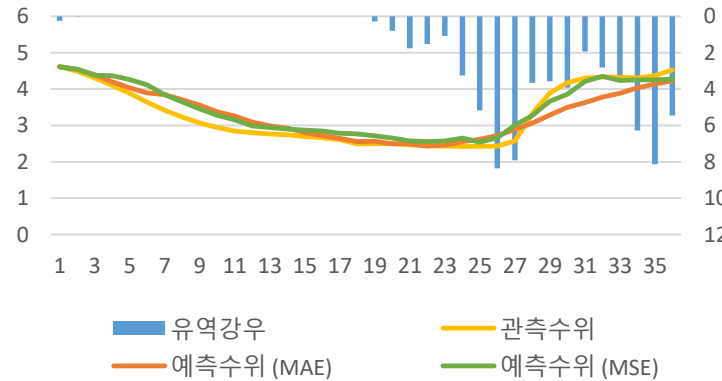
20230711 08시 대곡교



20230711 09시 대곡교

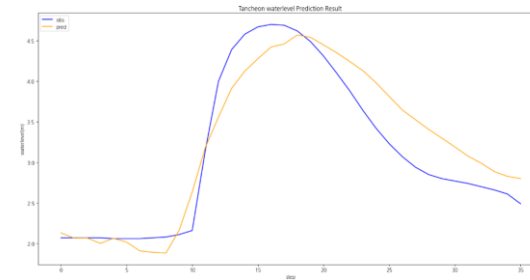


20230711 10시 대곡교

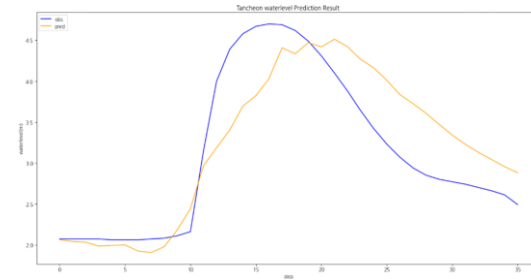


* R2(%) MSE(m²) MAE(m) RMSE(m)

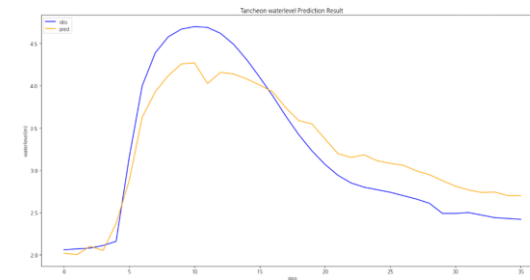
대곡교



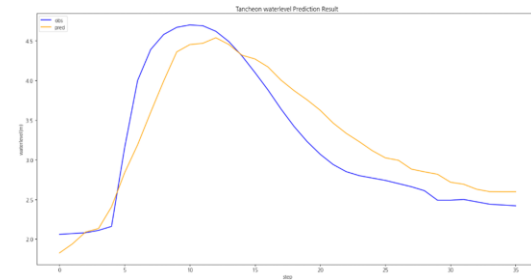
07_mse



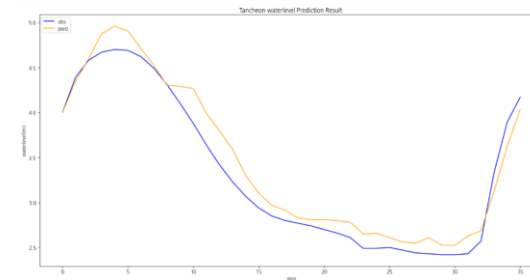
07_mae



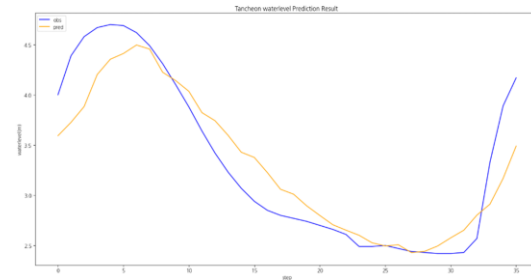
08_mse



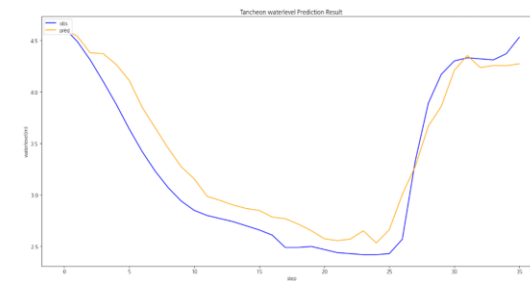
08_mae



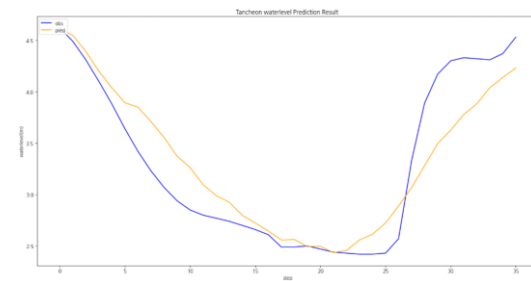
09_mse



09_mae



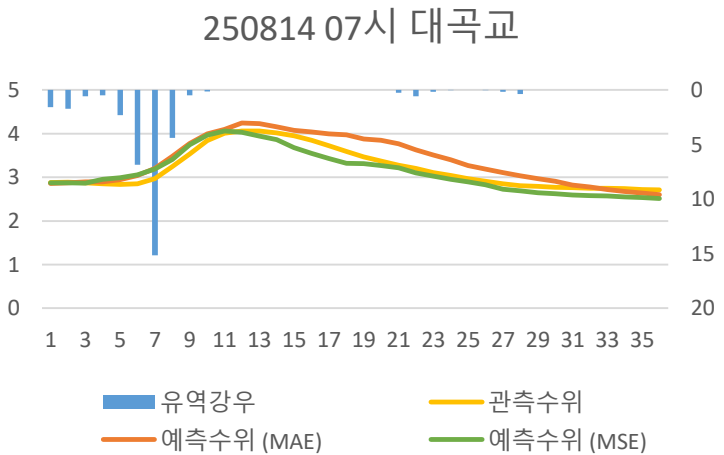
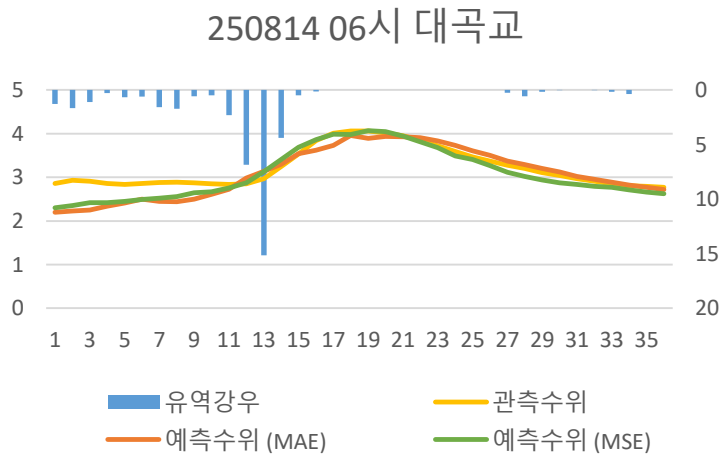
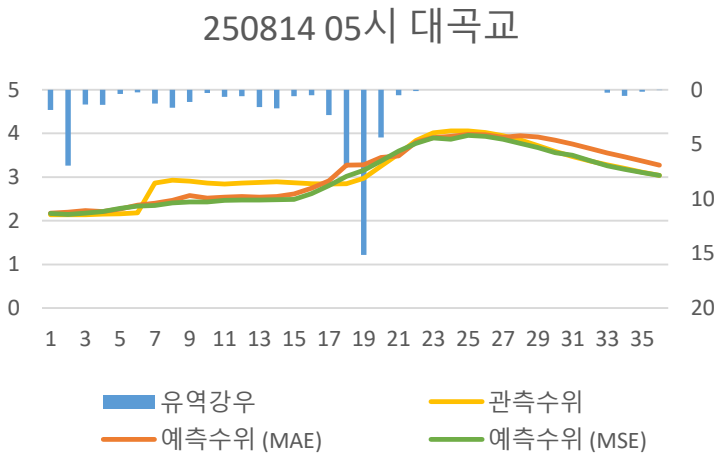
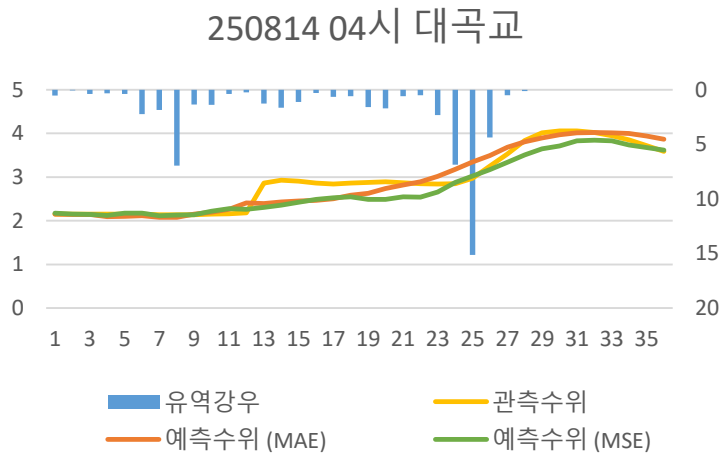
10_mse



10_mae

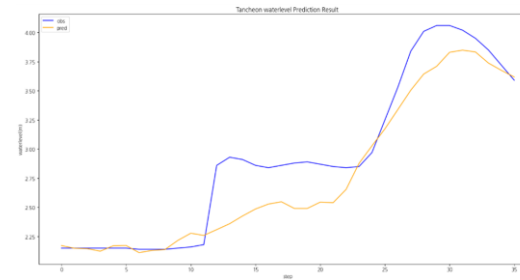
대곡교

20250814		MSE	MAE
04시	R2	0.86	0.90
	MSE	0.06	0.05
	MAE	0.19	0.16
	RMSE	0.25	0.22
05시	R2	0.84	0.83
	MSE	0.06	0.06
	MAE	0.17	0.21
	RMSE	0.24	0.24
06시	R2	0.71	0.59
	MSE	0.06	0.08
	MAE	0.18	0.20
	RMSE	0.24	0.28
07시	R2	0.88	0.73
	MSE	0.03	0.06
	MAE	0.14	0.20
	RMSE	0.16	0.24

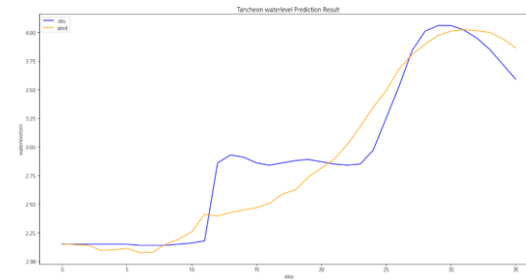


* R2(%) MSE(m²) MAE(m) RMSE(m)

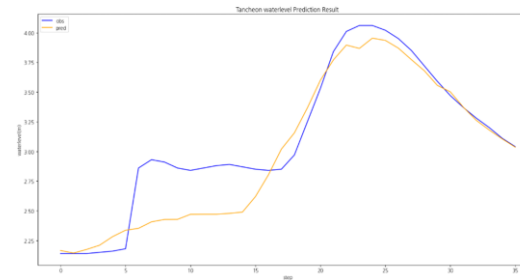
대곡교



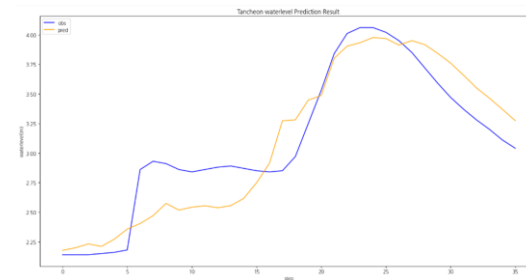
04_mse



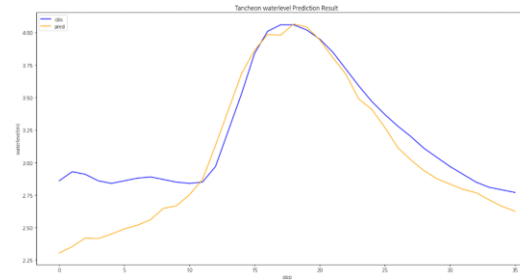
04_mae



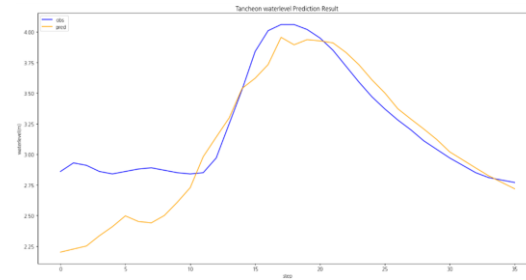
05_mse



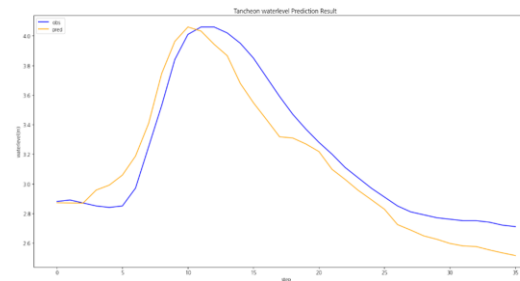
05_mae



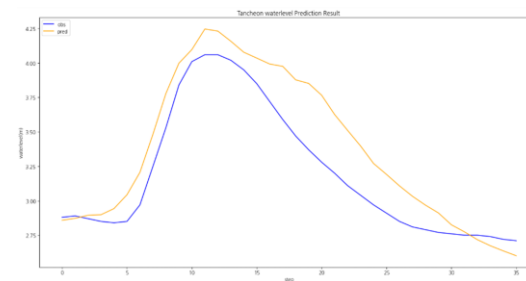
06_mse



06_mae



07_mse



07_mae